

MARINE COMMUNITY ECOLOGY II

SPECIATION



Assignment 2

Thursday's lab: more fish processing



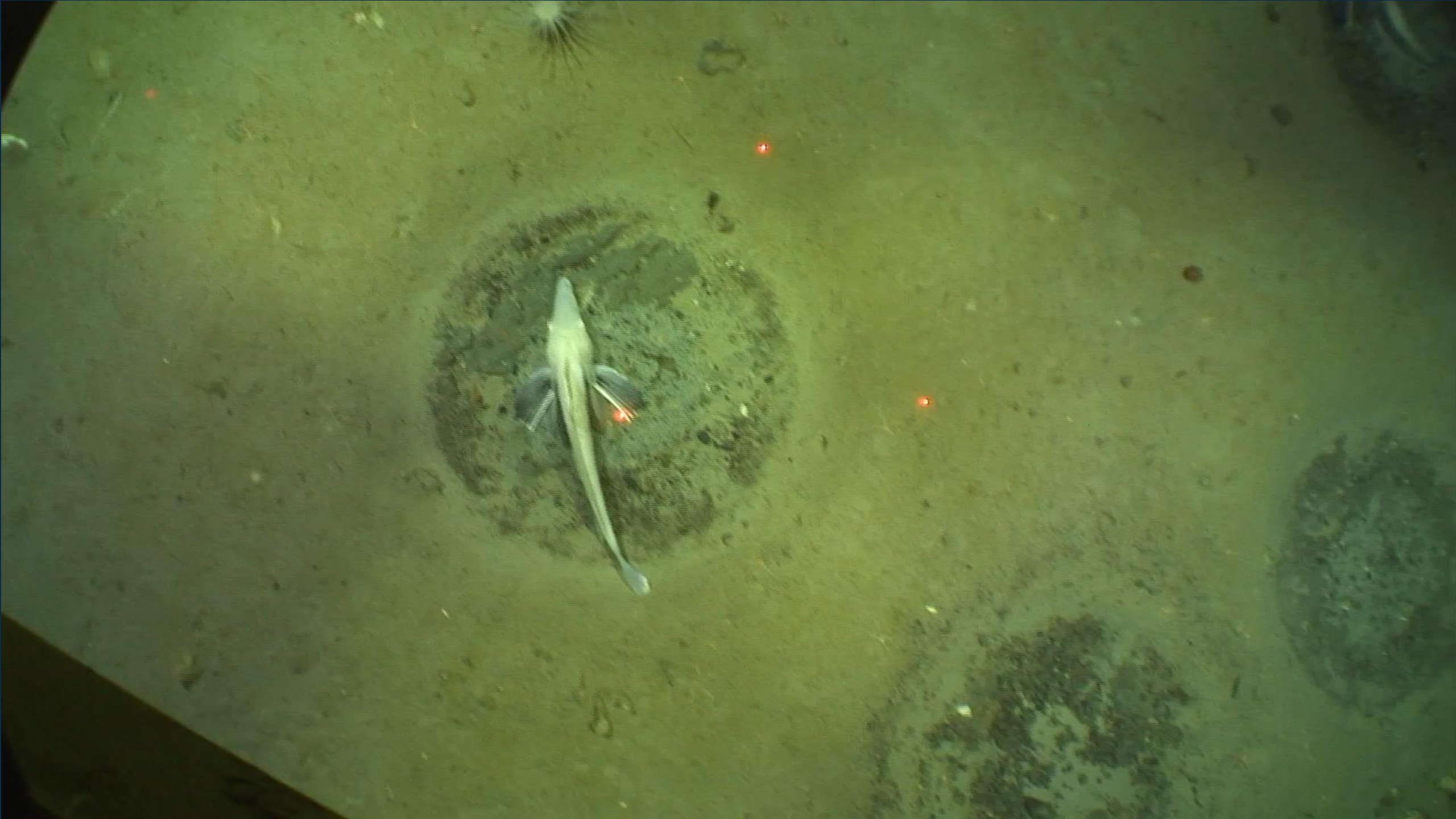
Channichthyidae: notothenioid icefishes







Filchner Trough



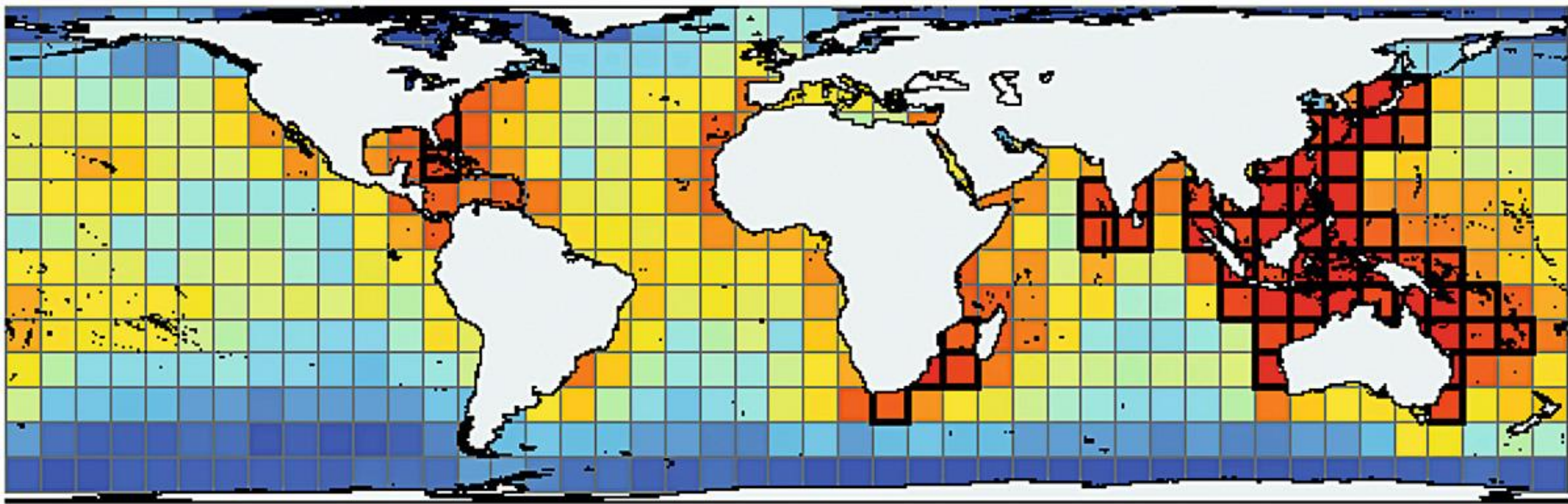


240km
1 nest every 4 m²
60,000 tons of fish biomass

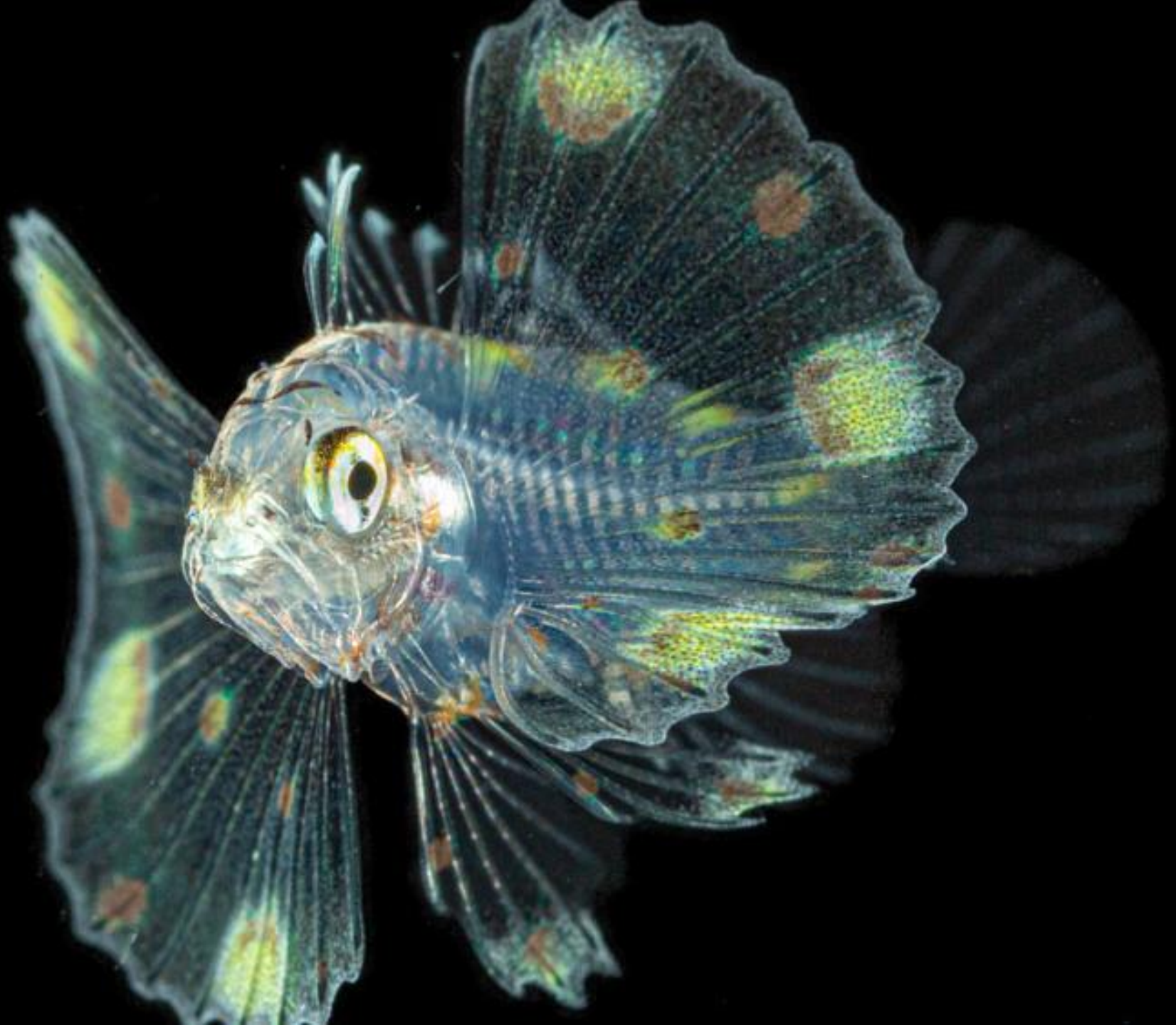
Just for the halibut...



Global biodiversity patterns



Dispersal



larval *Dendrochirus*

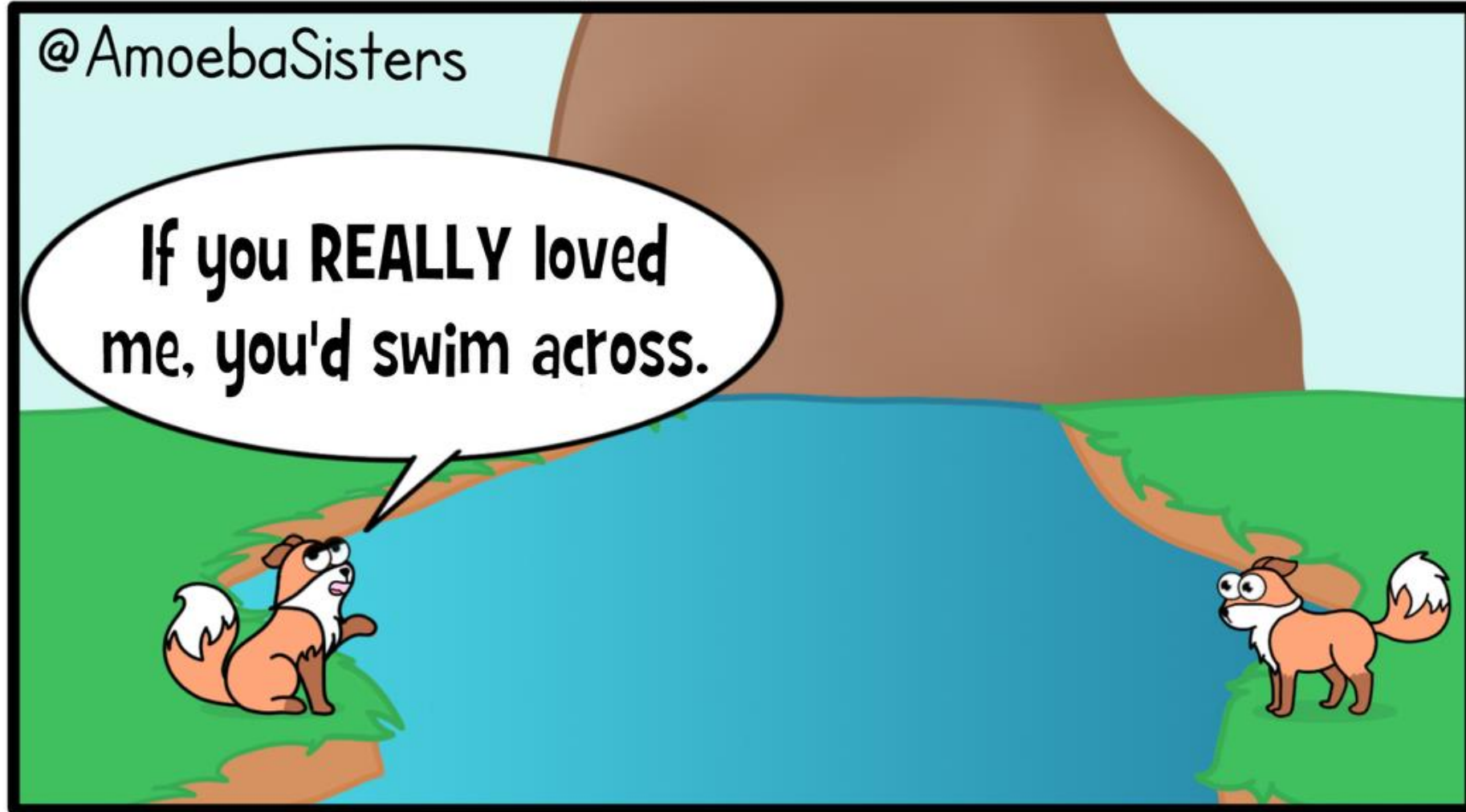


Speciation

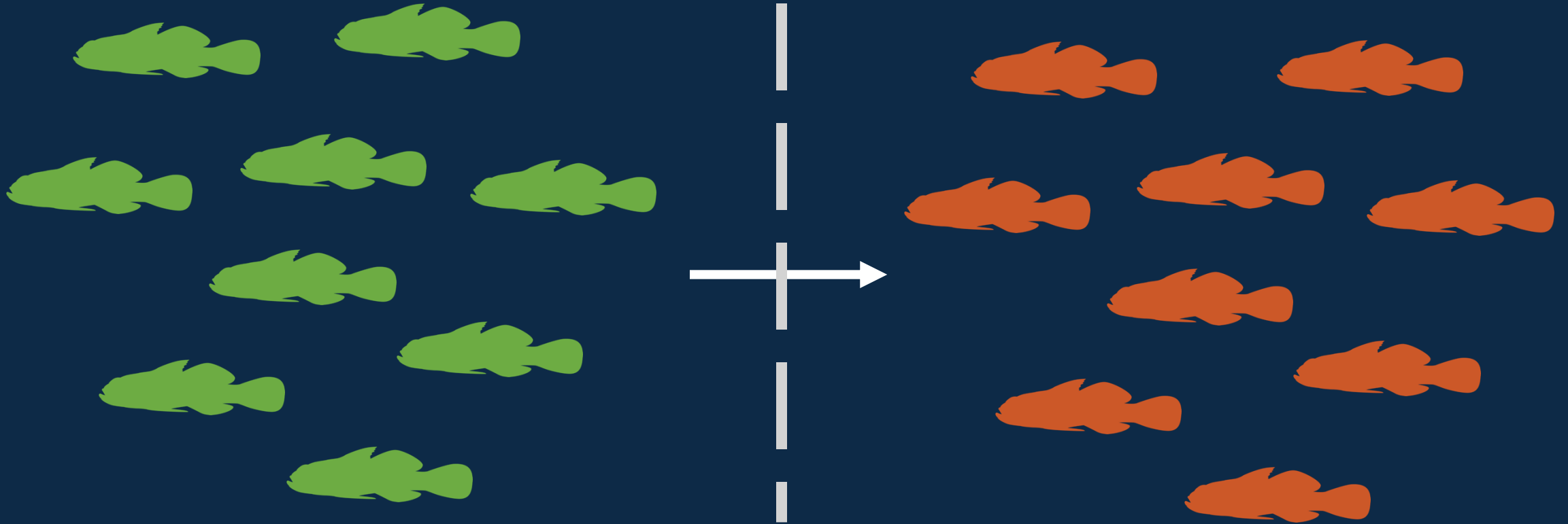
Pleurosicya mossambica

Reproductive isolation

Geographic Isolation



Allopatric speciation



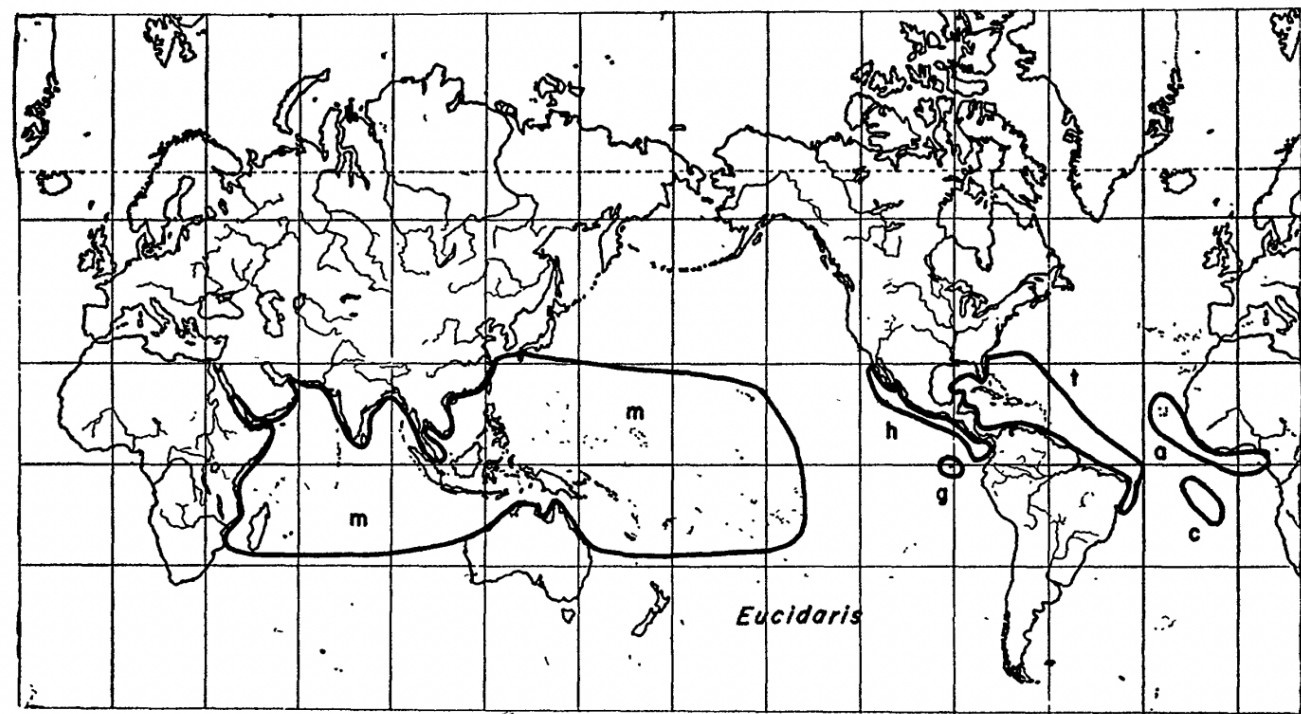
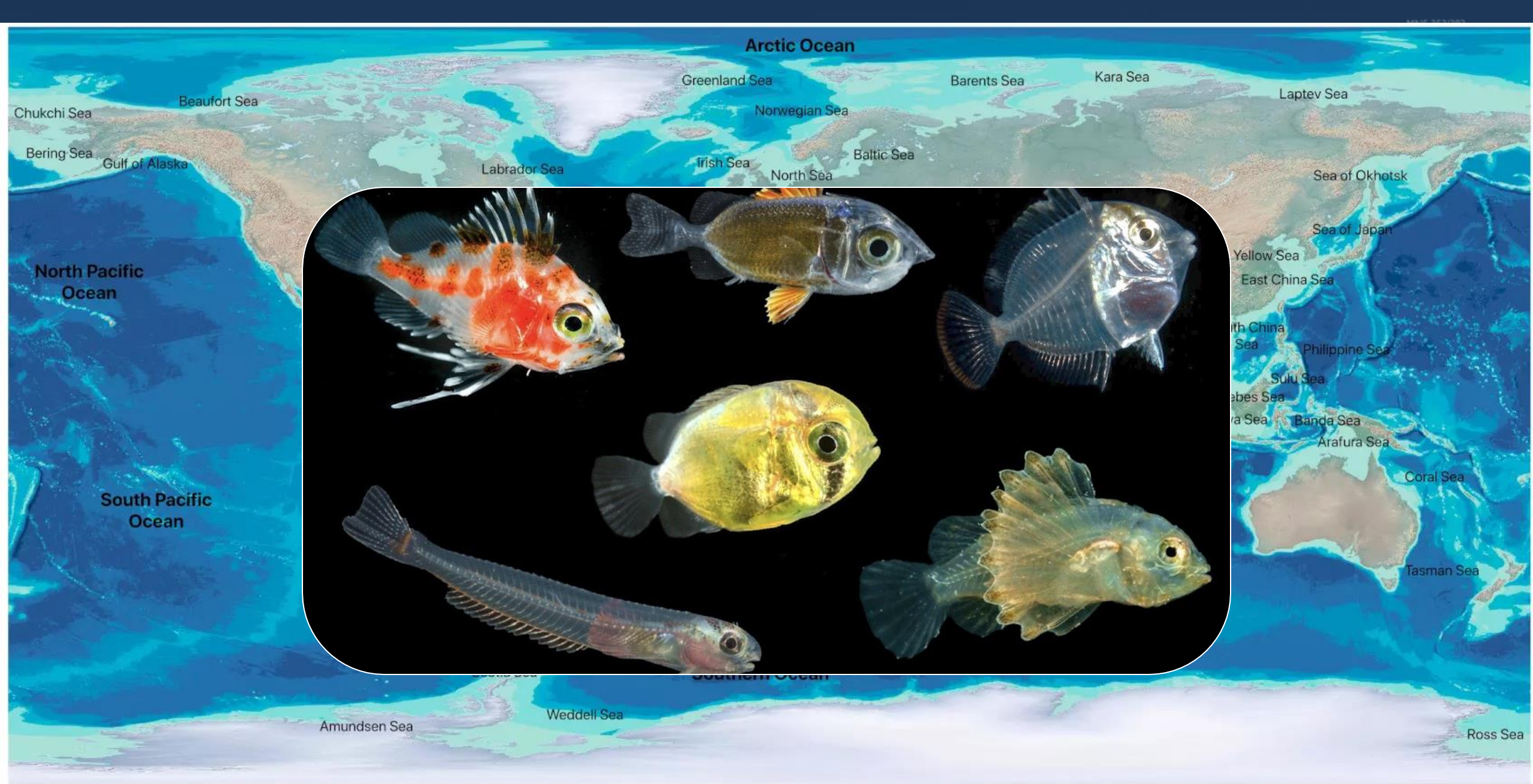


FIG. 1. Range of the species of *Eucidaris*. m = *metularia*, g = *galapagensis*, h = *thouarsii*, t = *tribuloides*, a = *africana*, c = *clavata*.



Biogeographic barriers in the oceans

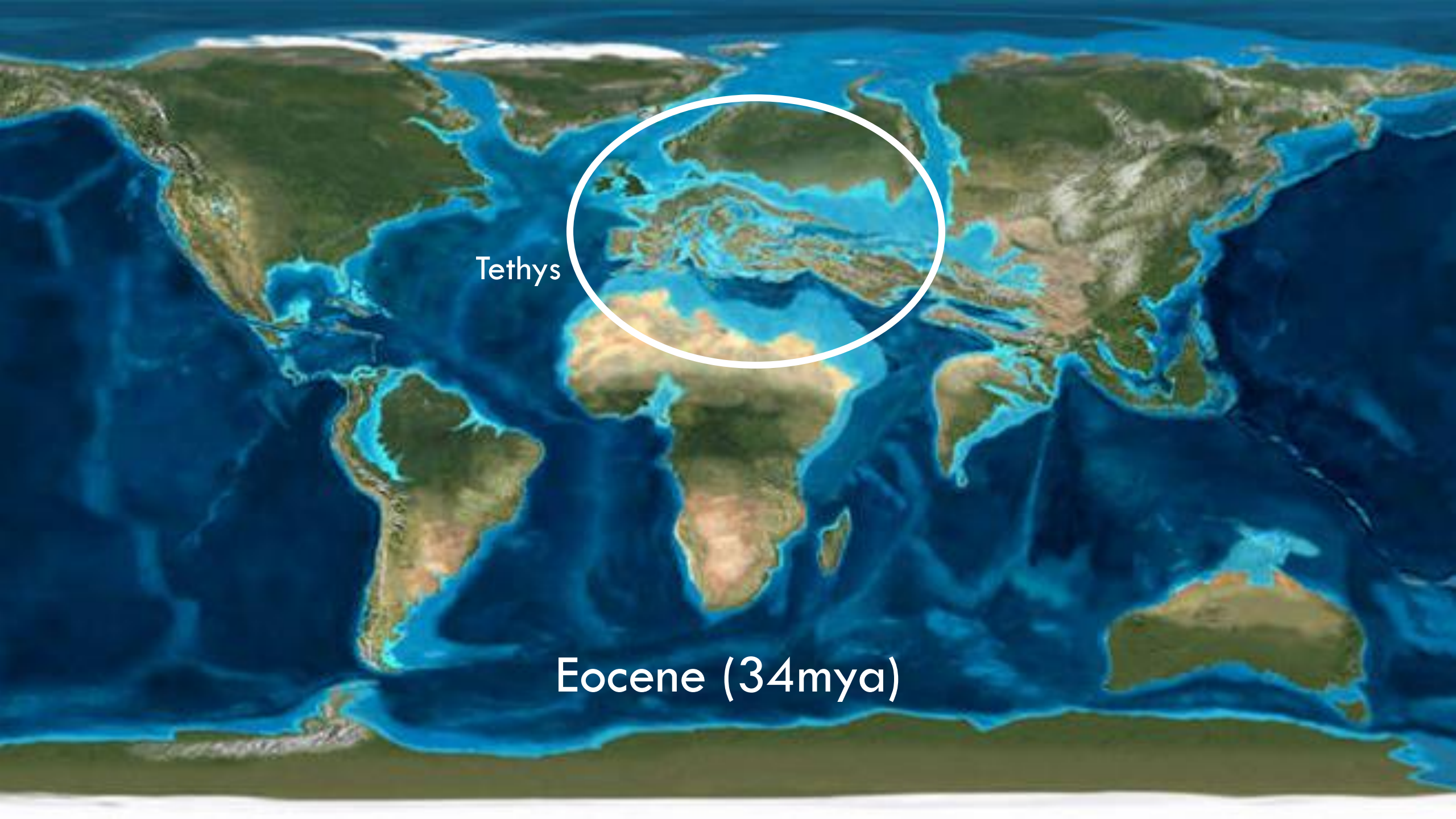
Hard barriers: Land bridges

Soft barriers: Water currents, distance

IF SPECIES SPECIATED...

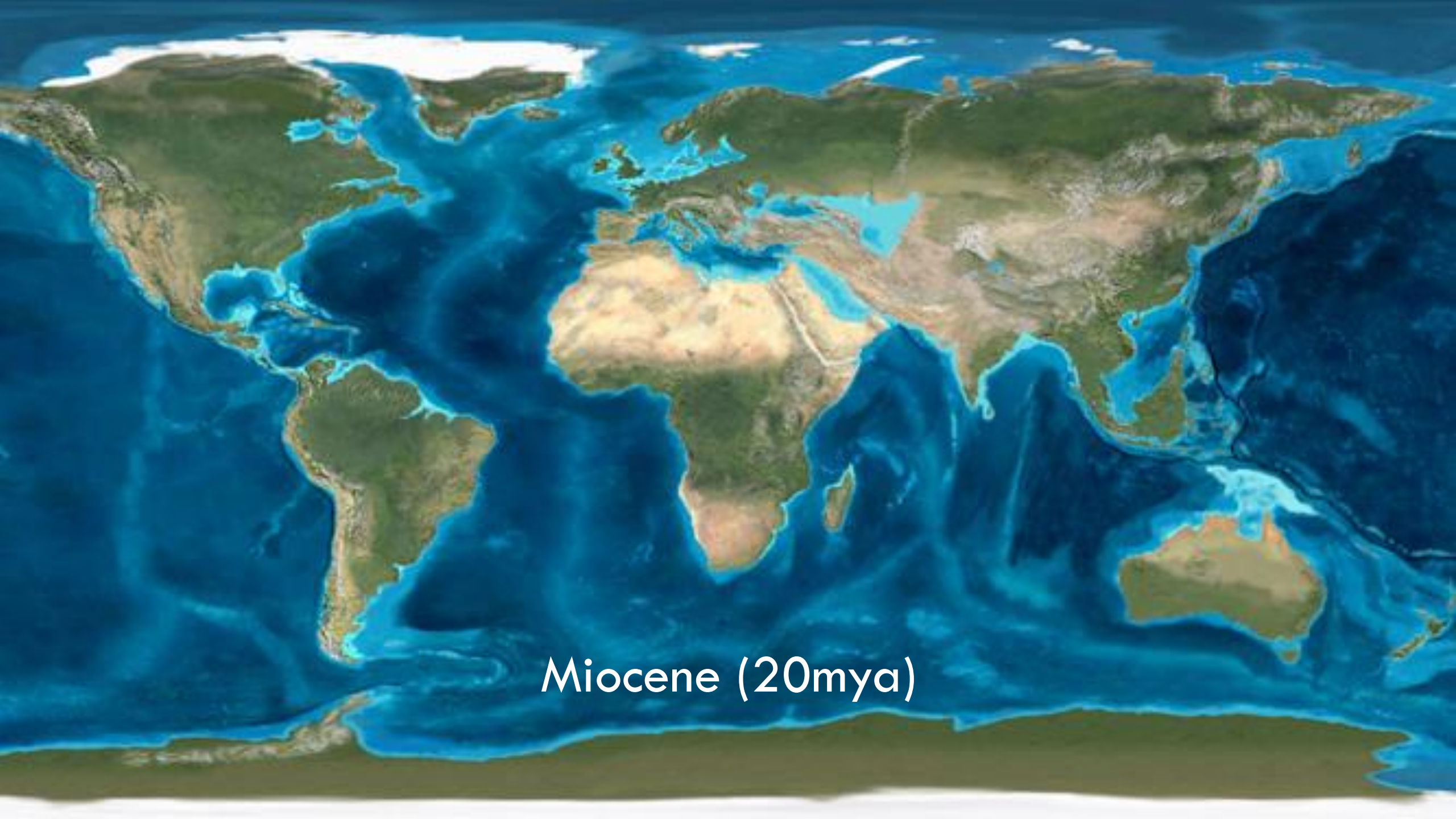


DID GENERA GENERATE?

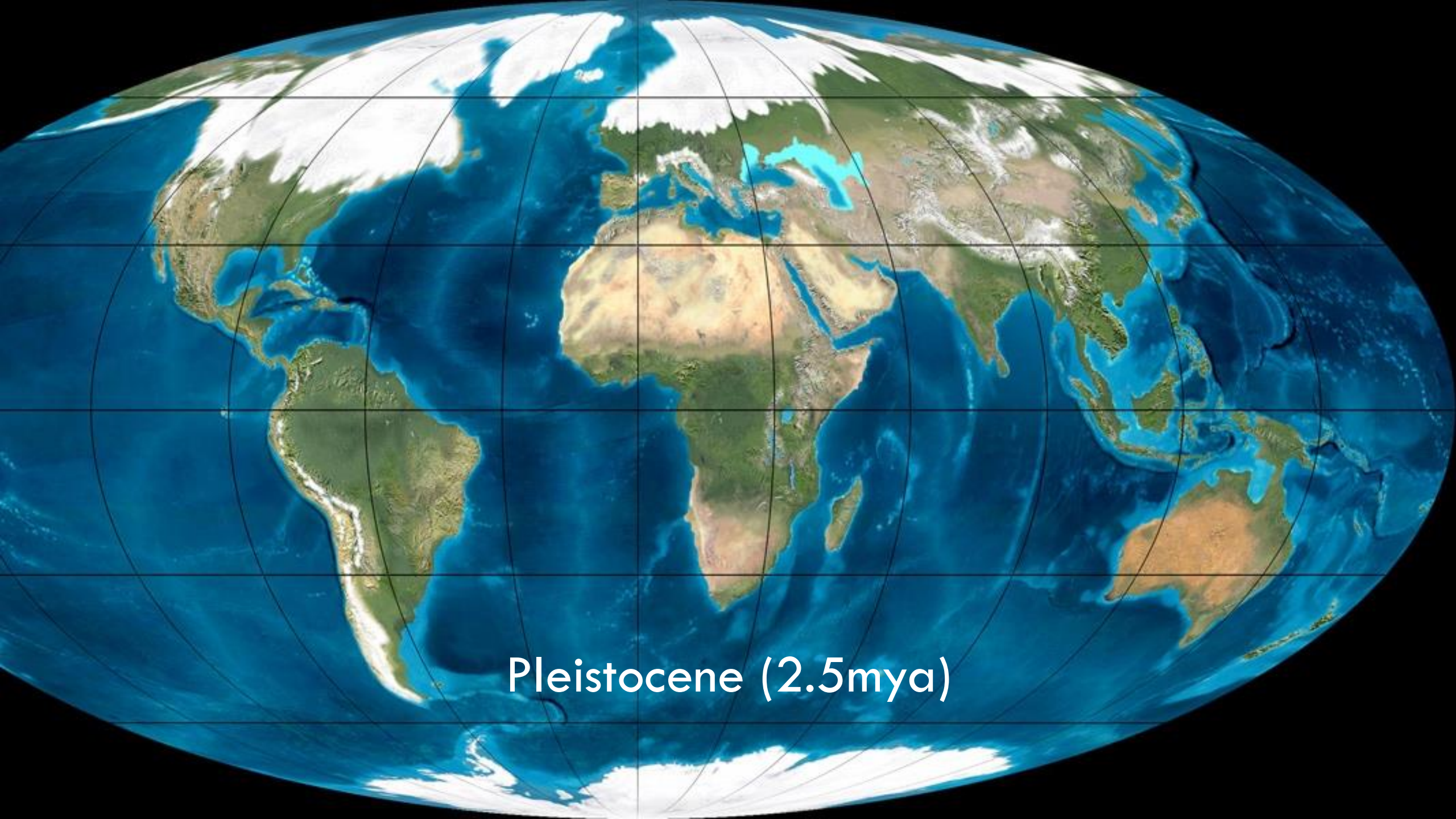


Tethys

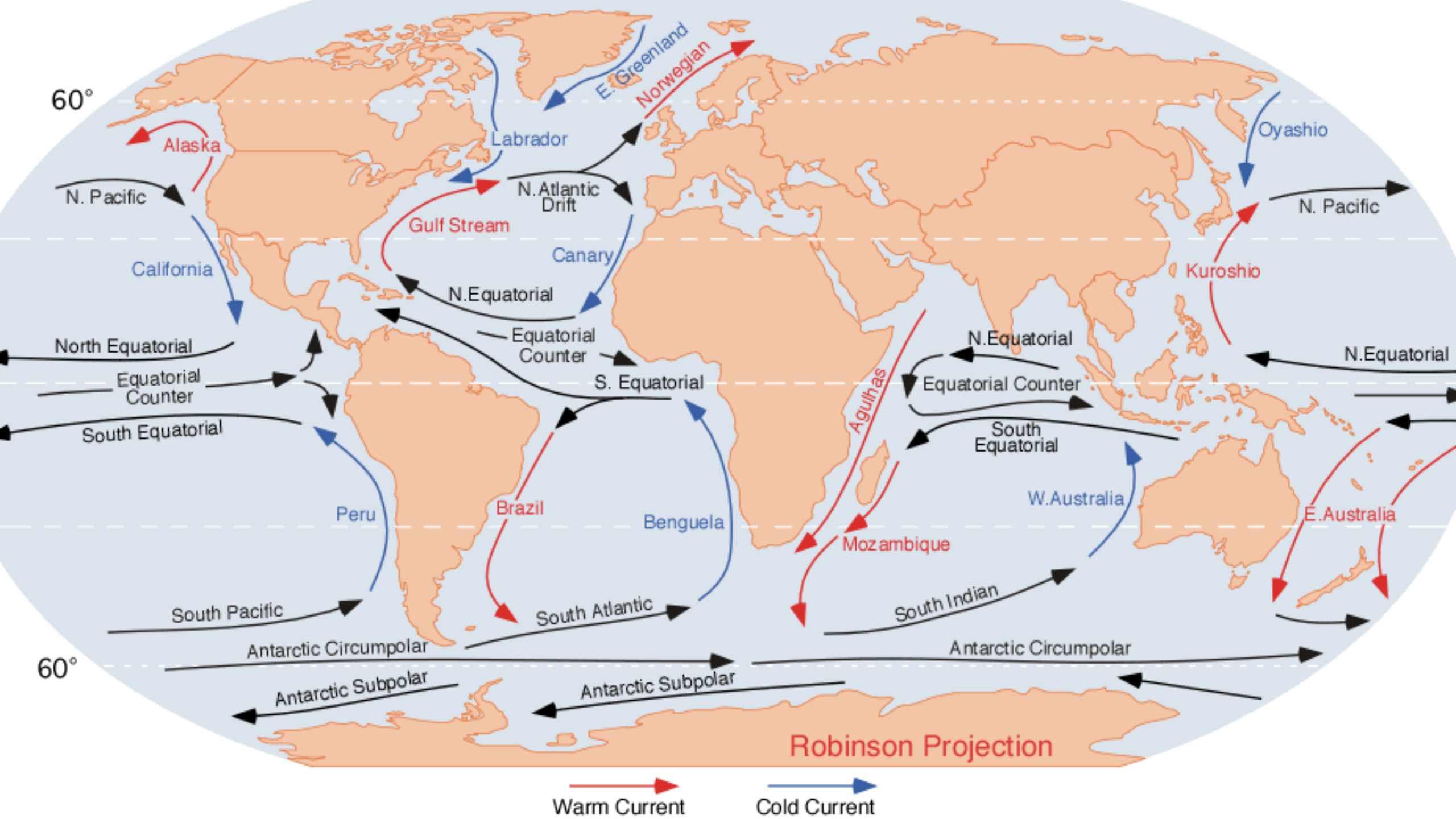
Eocene (34mya)



Miocene (20mya)



Pleistocene (2.5mya)

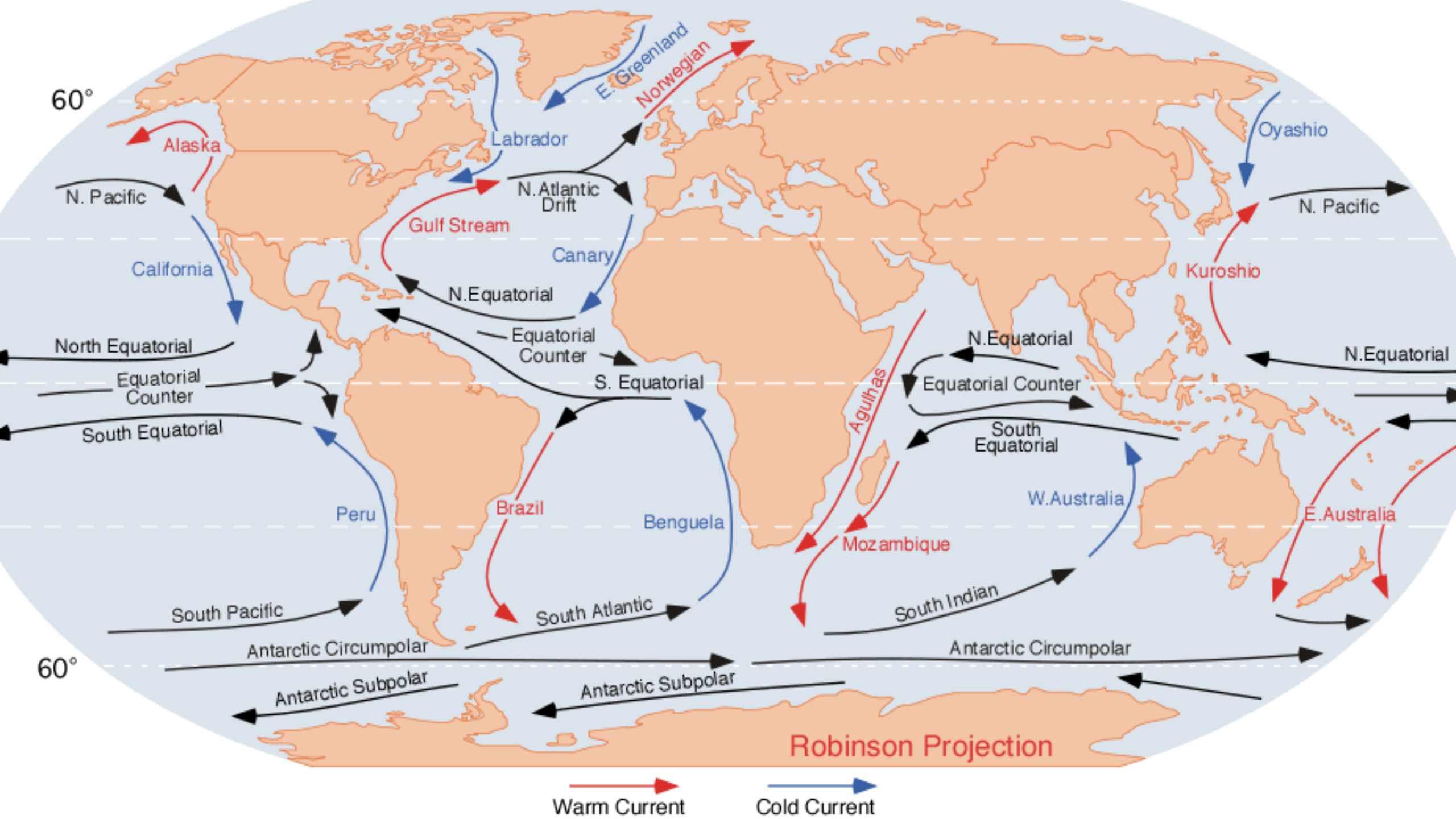


Discussion

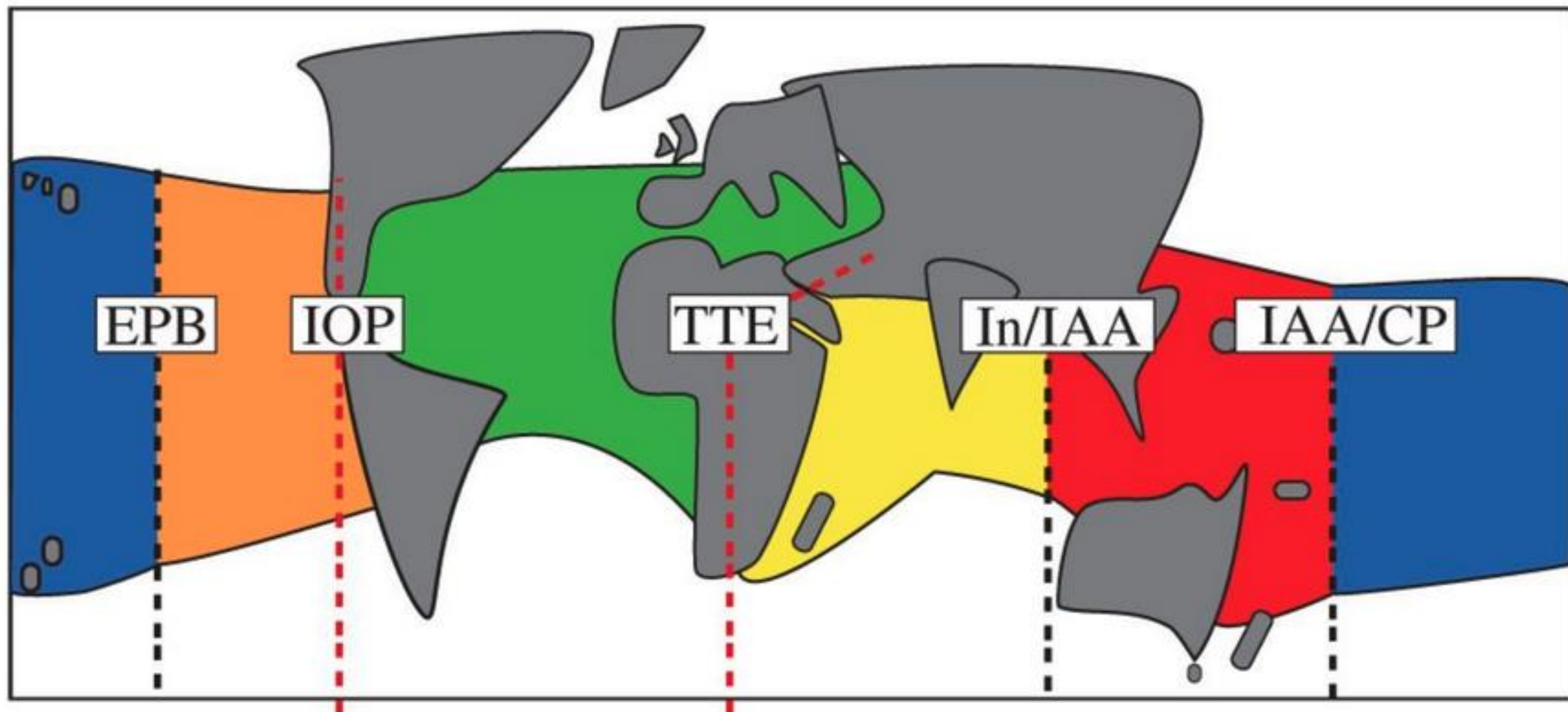
Take 5 minutes to discuss:

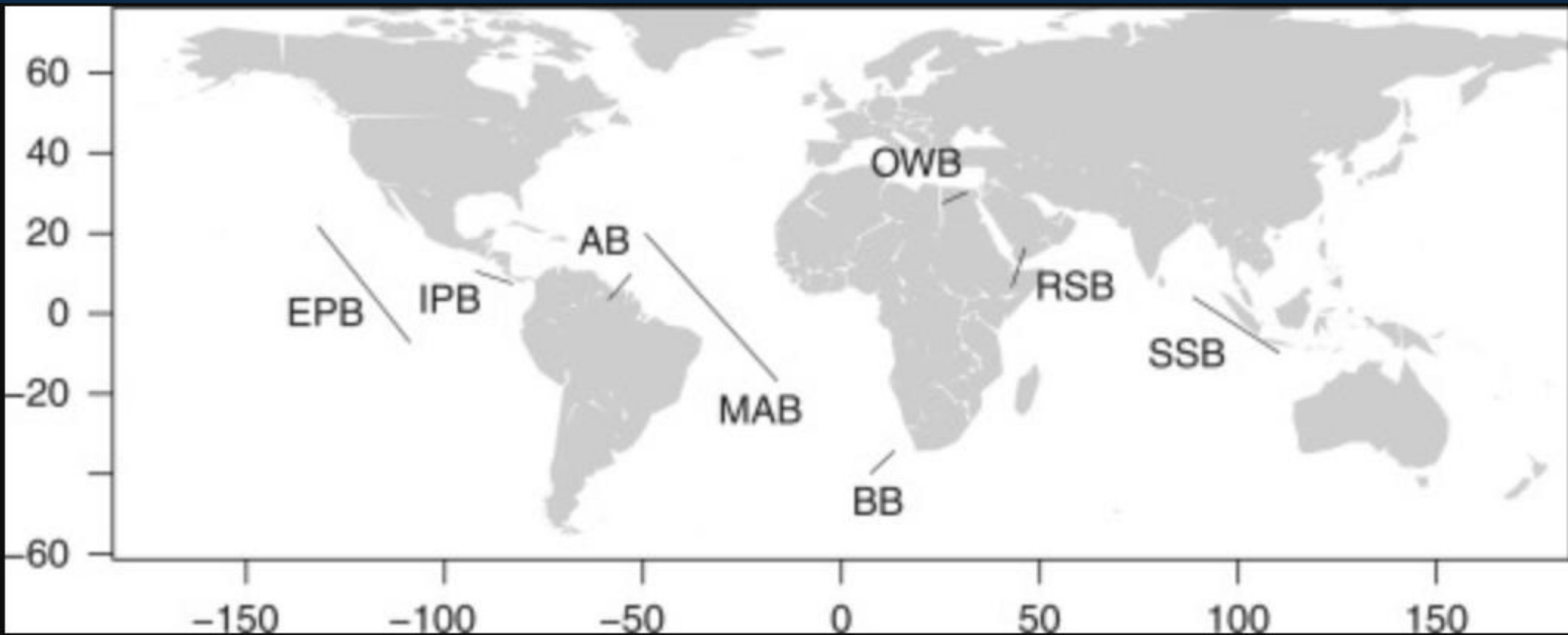
- 1) What are the major biogeographic barriers in the oceans and are they hard or soft barriers (8 total)?
- 2) Are the barriers different for tropical versus temperate versus polar species?





(a)





Major barriers in marine environments

Terminal Tethyan Event (TTE, 12-18mya) - hard barrier

Isthmus of Panama (IOP, 3mya) - hard barrier

Eastern Pacific Boundary (EPB, 65 mya) - soft barrier

Indian Ocean/Indo-Australian Archipelago - soft barrier

Indo-Australian Archipelago/Central Pacific - soft barrier

Amazon River Plume - soft barrier

Benguela current - soft barrier

Mid Atlantic Barrier - soft barrier

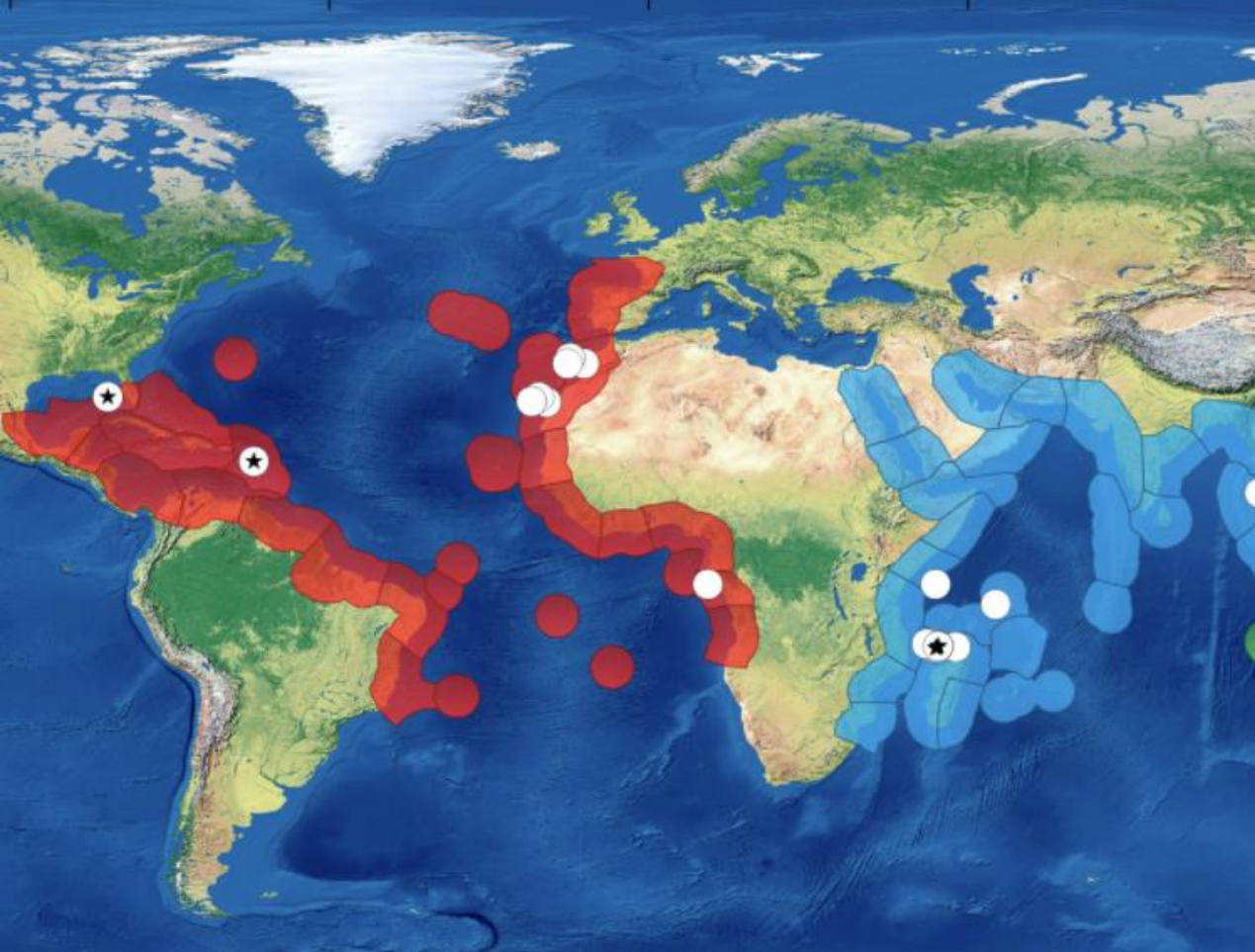
Red Sea Barrier - soft barrier

ASSIGNMENT 2

UNDERGRADS



Terminal Tethyan Event (12-18mya)



Leiolgalathea laevirostris (Balss, 1913)
光額滑鎧蝦



Fig. 113. Male (4.6 mm), DW45, body orangish.



Thalassoma wrasses





Great American Biotic Interchange (GABI)



Marine organisms be like...



Echinometra vanburen



Echinometra viridis



Alpheus rostratus



Alpheus paracrinitus



Callinectes toxotes



Callinectes sapidus





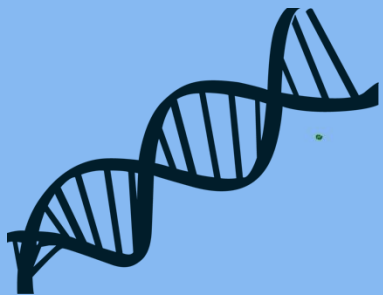
Pomacanthus zonipectus

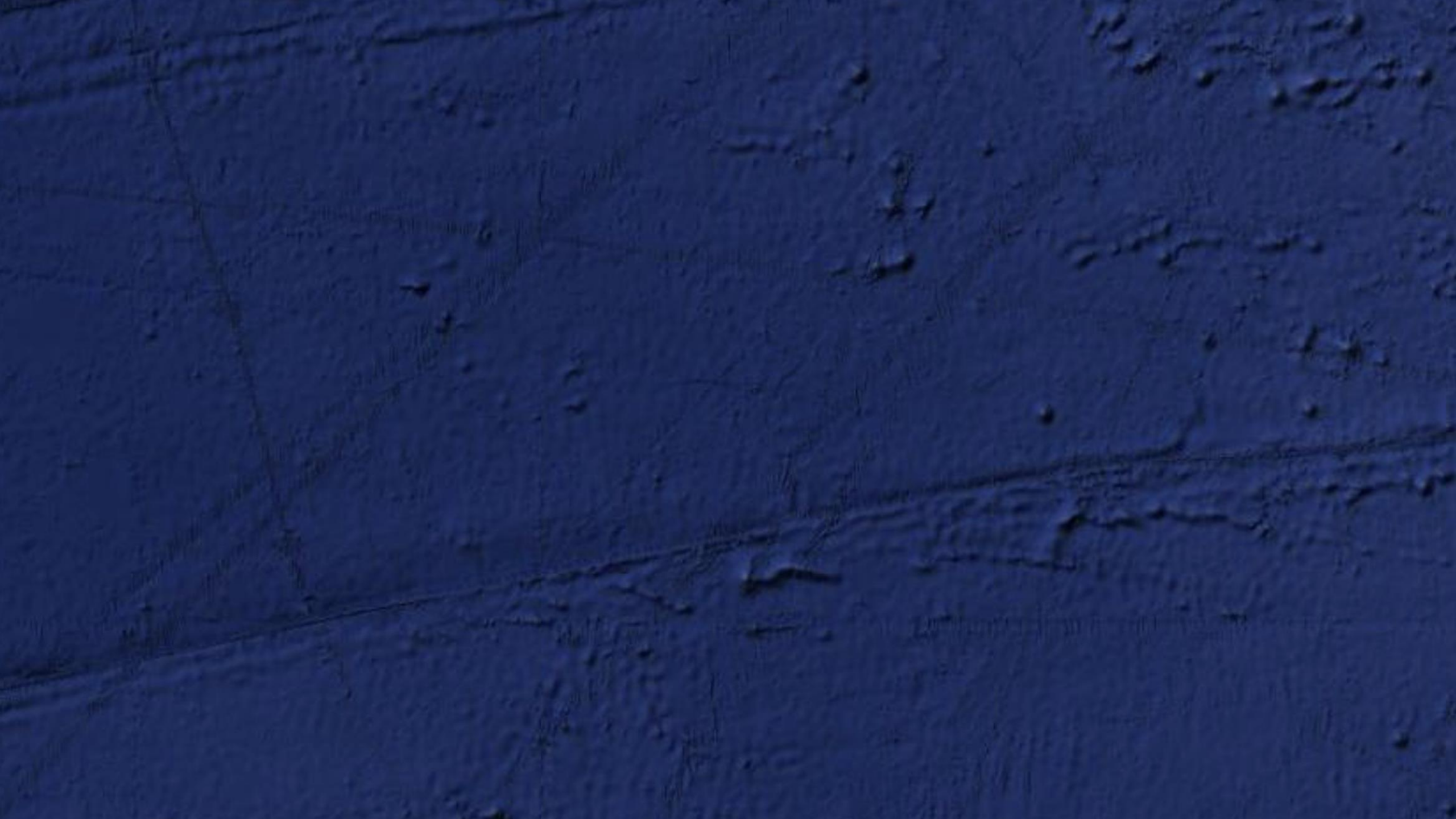


Pomacanthus paru



9 clades of echinoids
38 clades of crustaceans
42 clades of fishes
26 clades of molluscs





East Pacific Barrier (65mya)

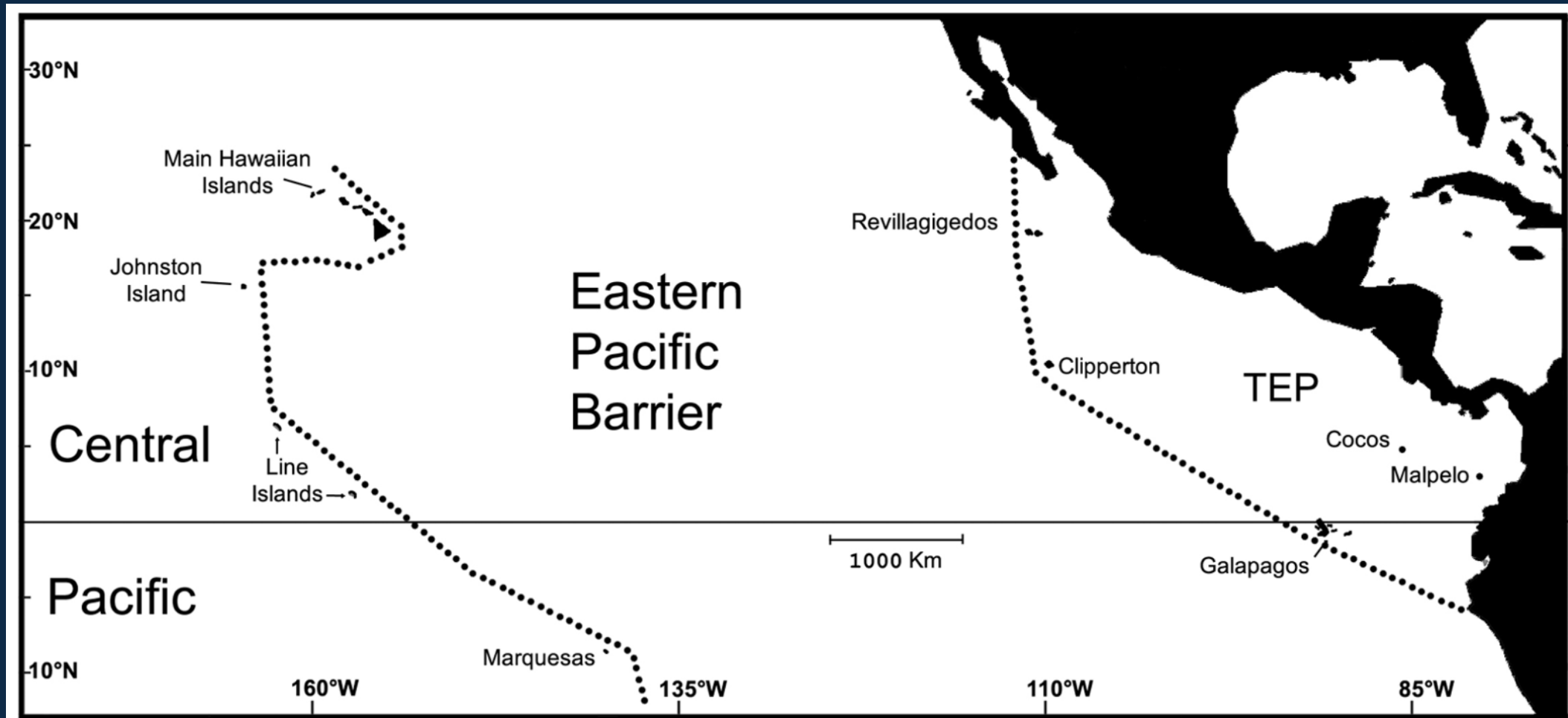


FIGURE 1. The tropical eastern Pacific (TEP) biogeographic region and its isolation from other regions by the Eastern Pacific Barrier and the Central American isthmus.





Plagiotremus azaleus



Prionurus punctatus



Kyphosus azurea



Synchiropus atrilabiatus

Chaenopsidae: genus *Acanthemblemaria*



© Carlos & Allison Estape, <http://carlosestape.com>



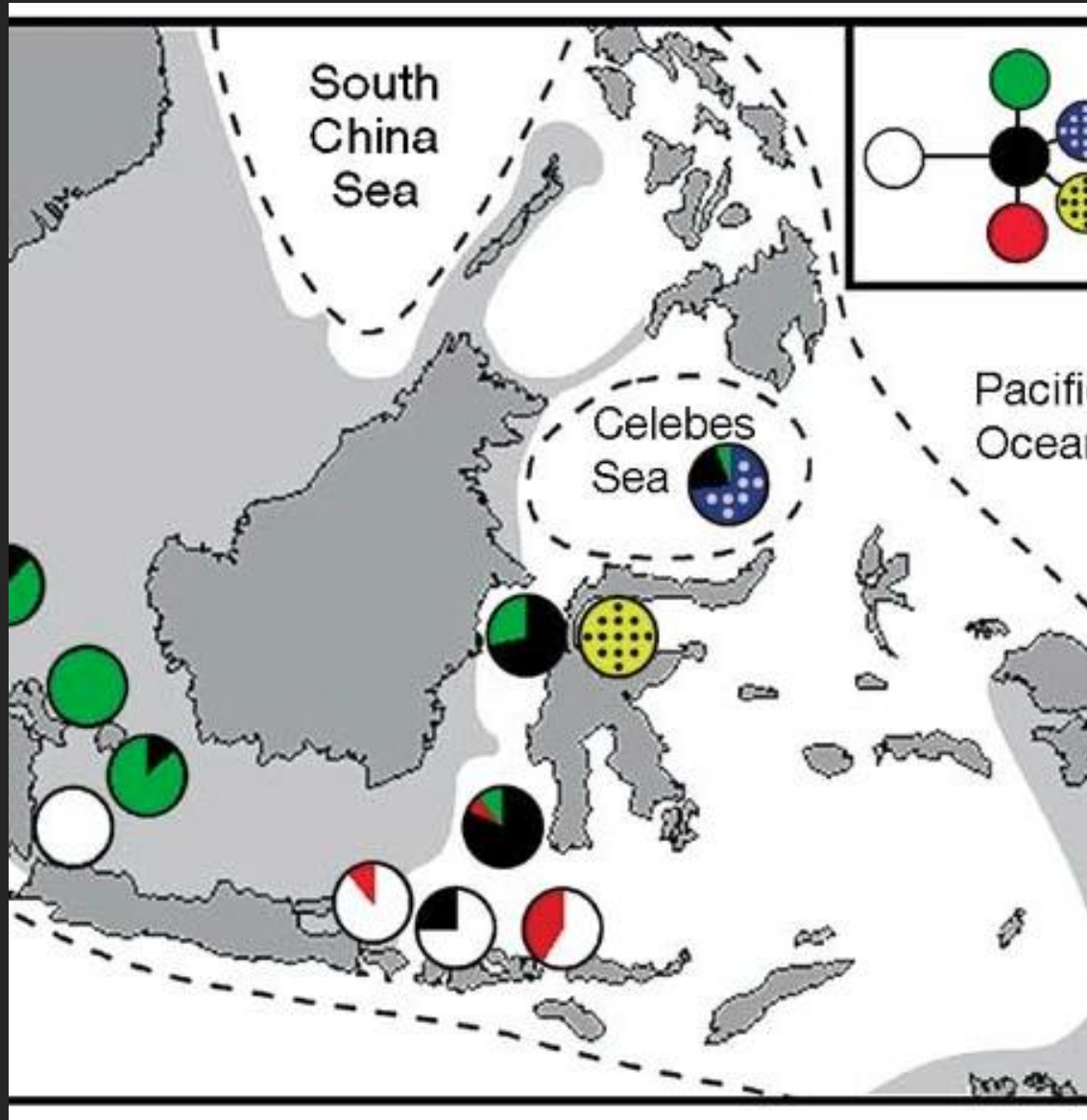


Pliocene-Pleistocene glaciations ($\sim 3\text{mya}$)

Indian Ocean/Indo-Australian Archipelago

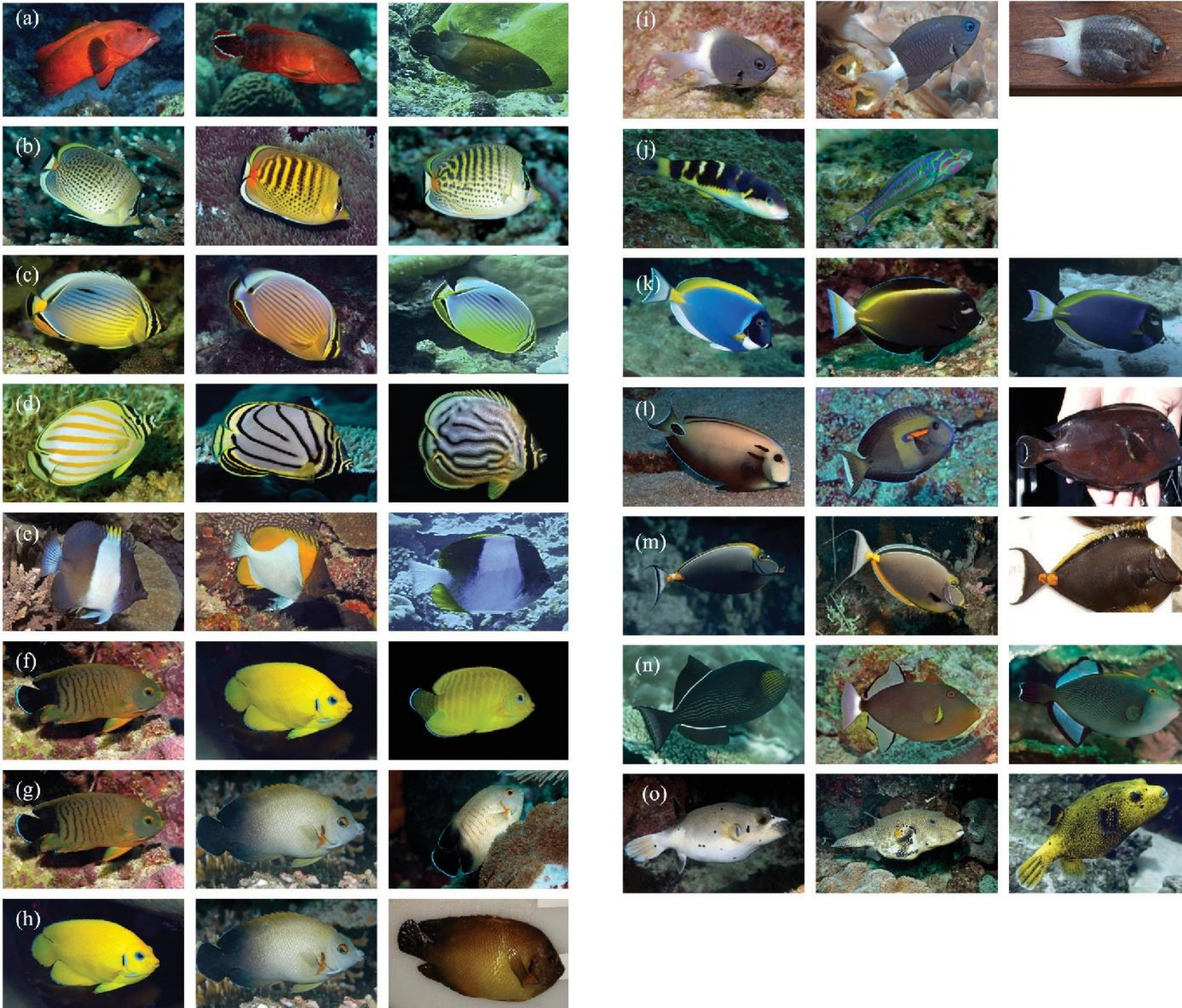
Wallace Line











Hobbs & Allen 2014

(a)



(b)



(c)



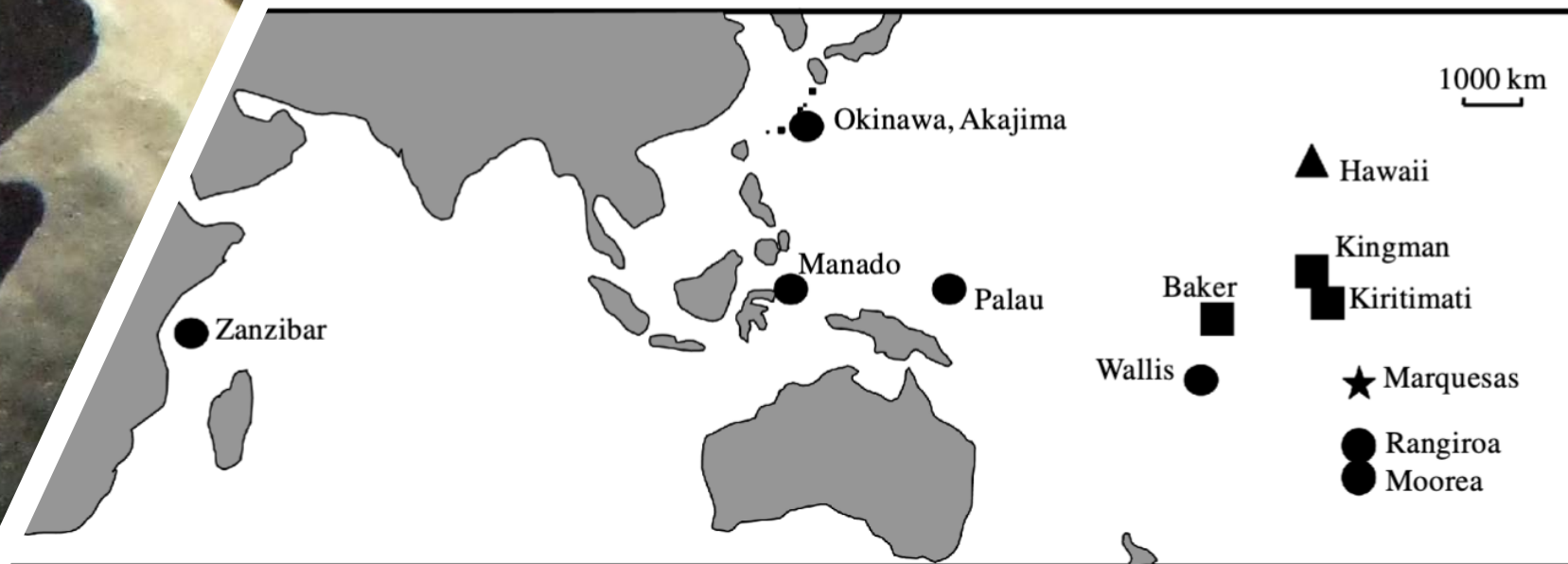
(d)



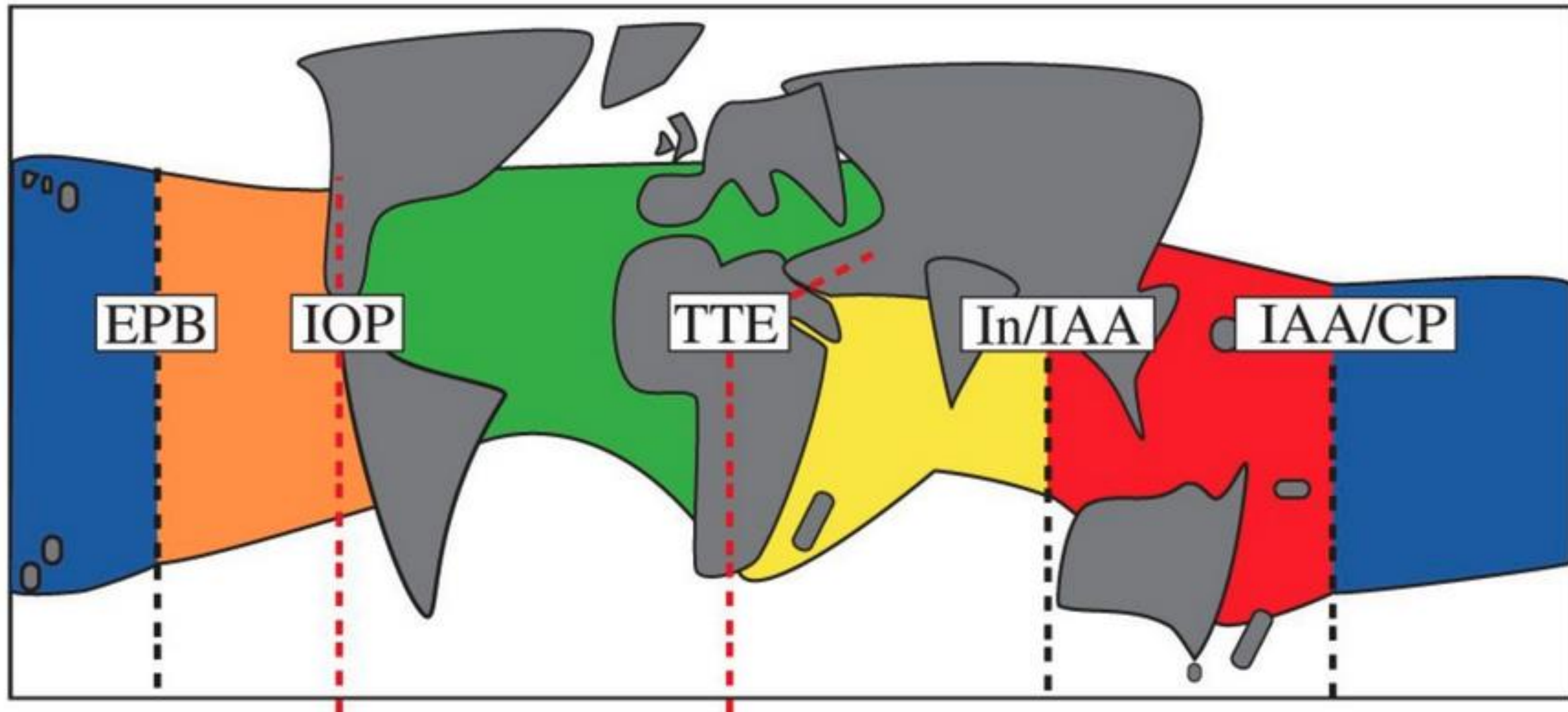


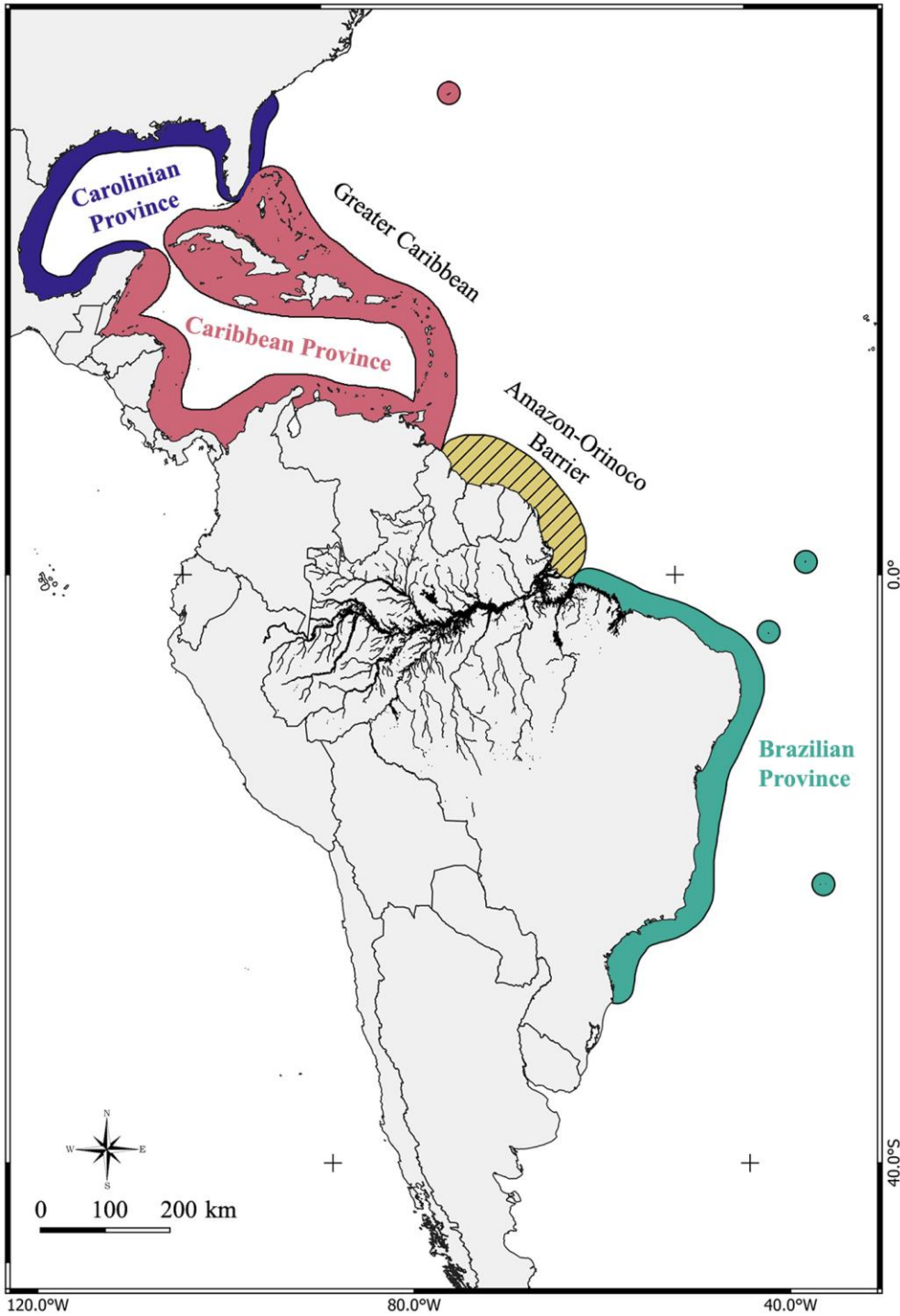
Indo-Australian Archipelago/South Pacific





(a)

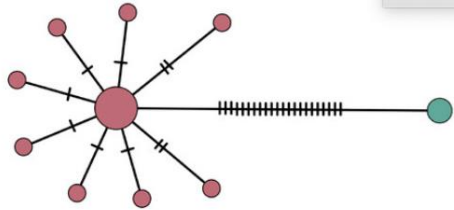
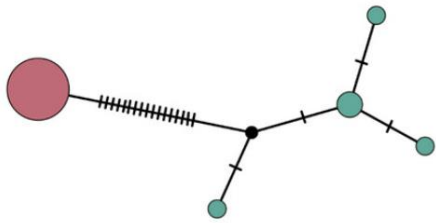
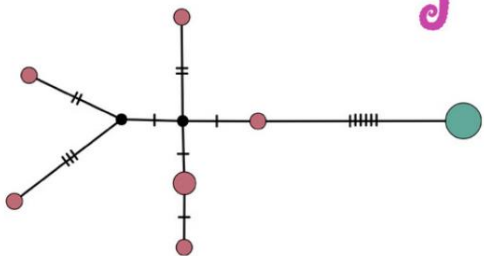
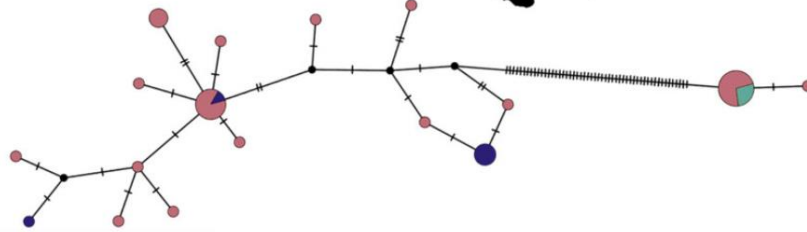
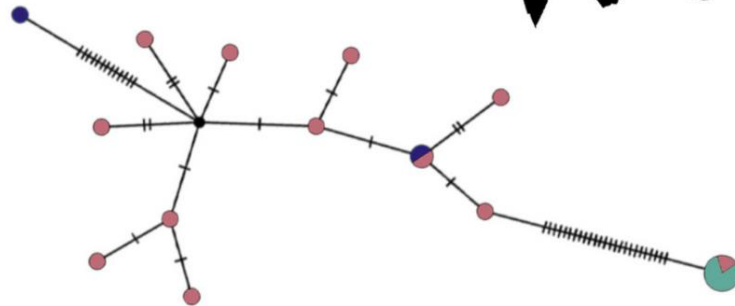
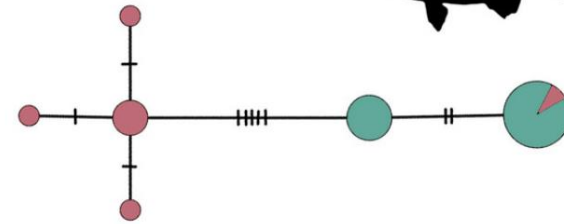
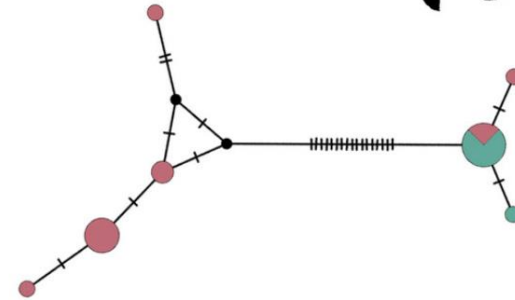
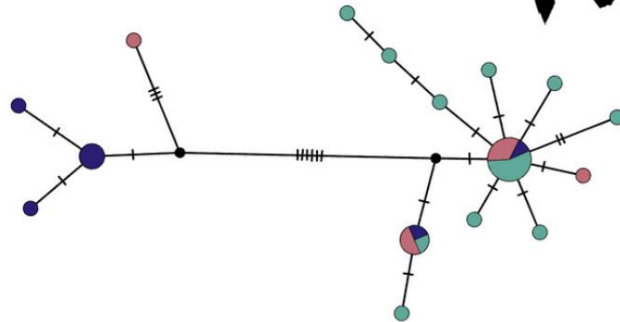
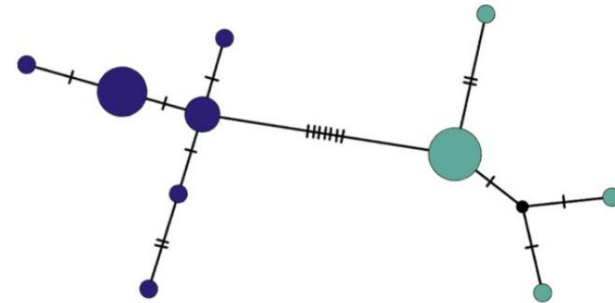




(a)

Gobiesox barbatulus*Coryphopterus thrix*

Details are in the caption following the image

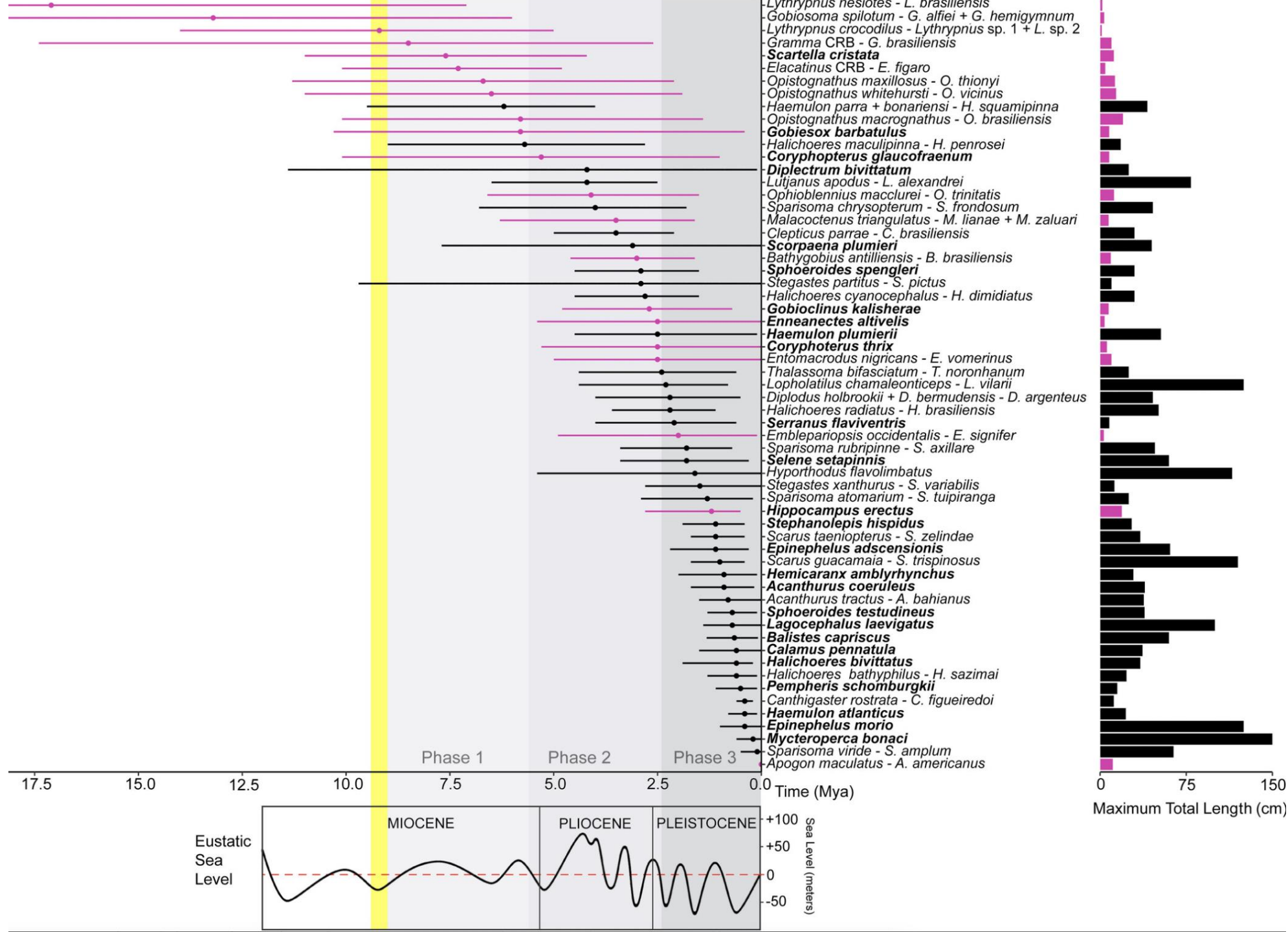
*Enneanectes altivelis**Hippocampus erectus**Scorpaena plumieri**Haemulon plumierii**Haemulon atlanticus**Diplectrum bivittatum**Haemulon aurolineatum**Selene setapinnis*

Caribbean Province

Brazilian Province

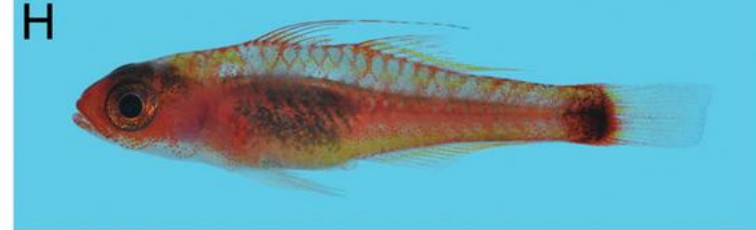
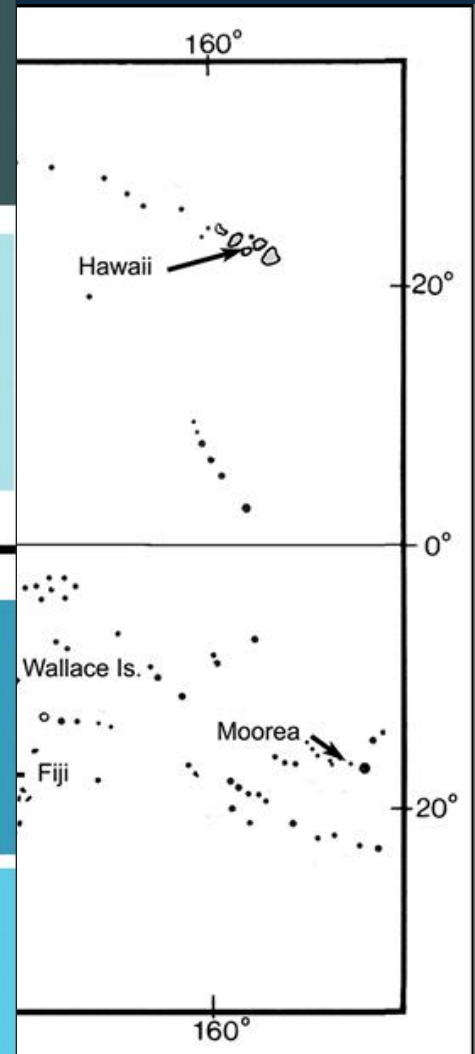
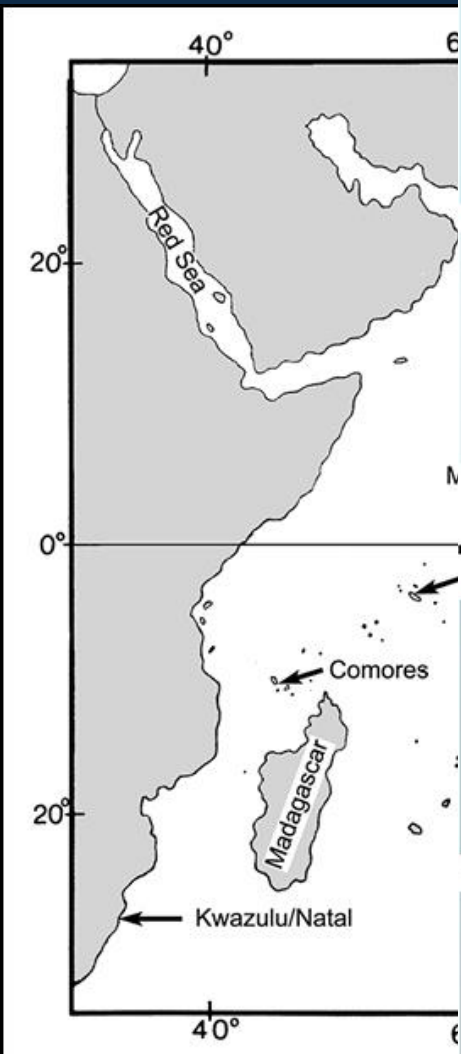
Carolinian Province

(c)

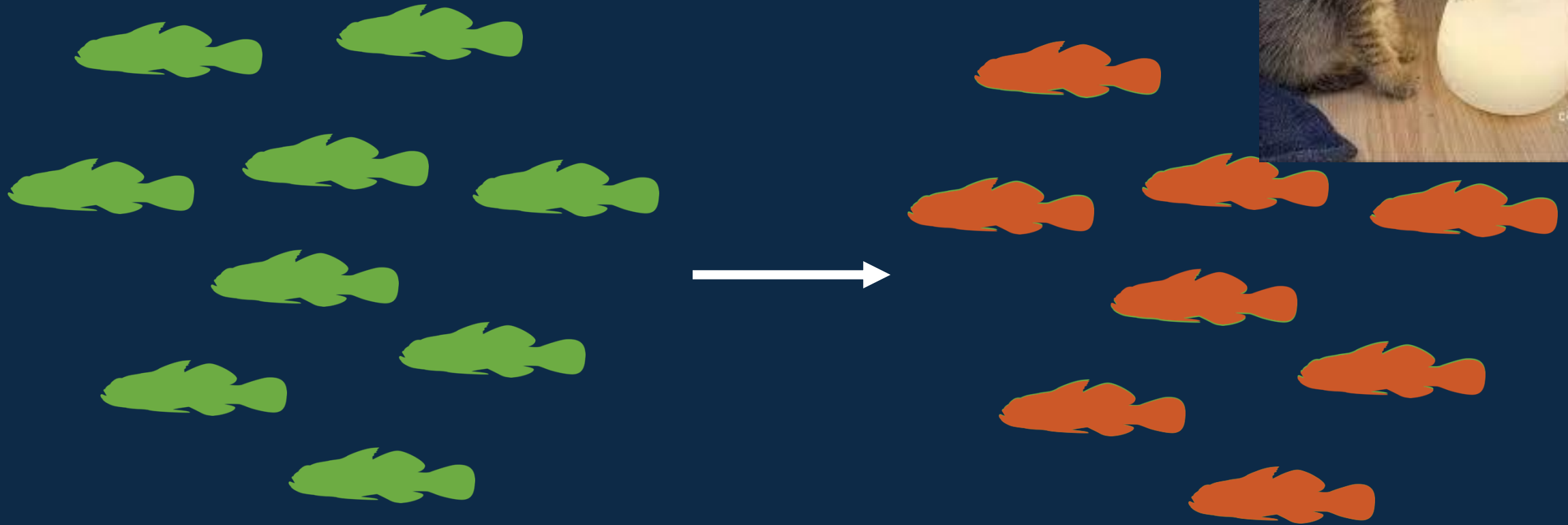


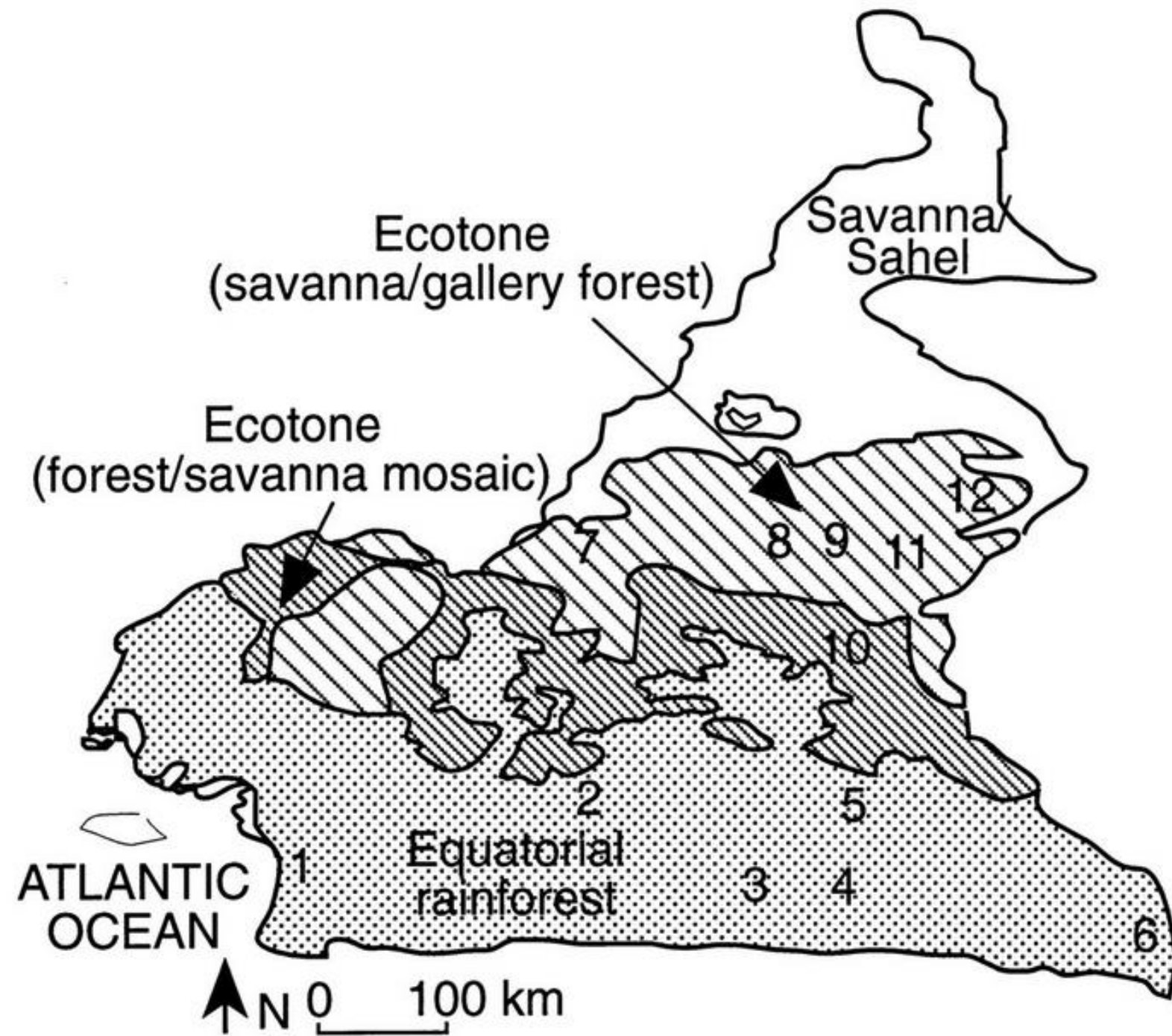






Sympatric speciation





Smith et al. 1997, Science

Allopatric speciation



Ecological speciation

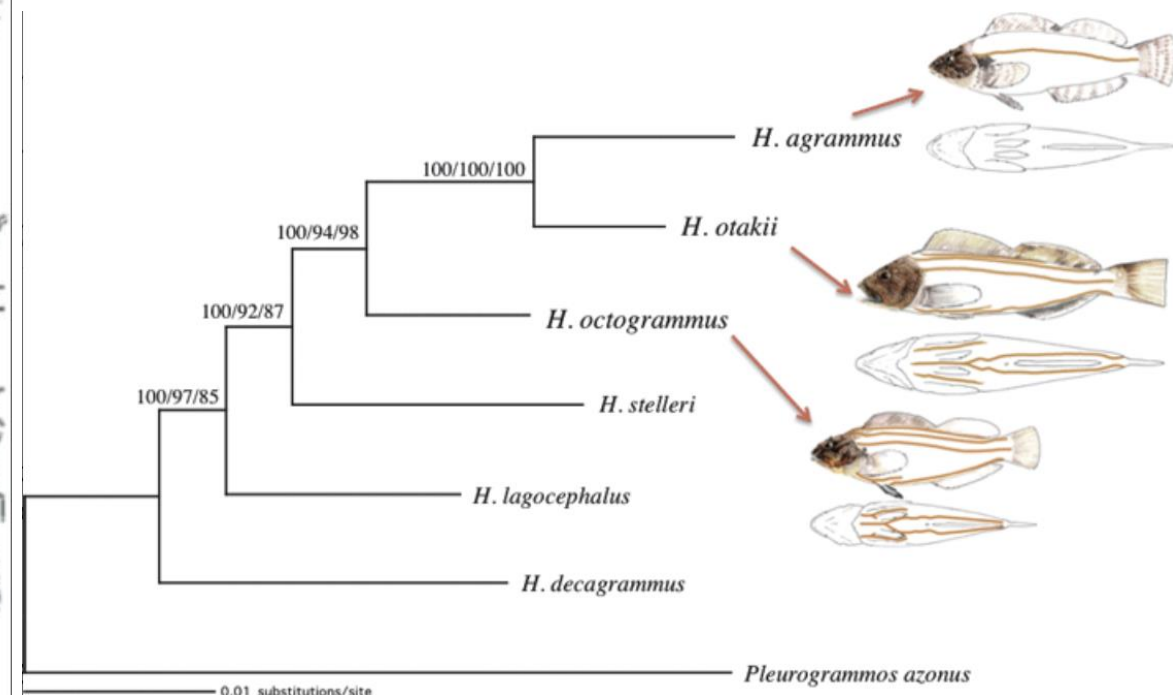
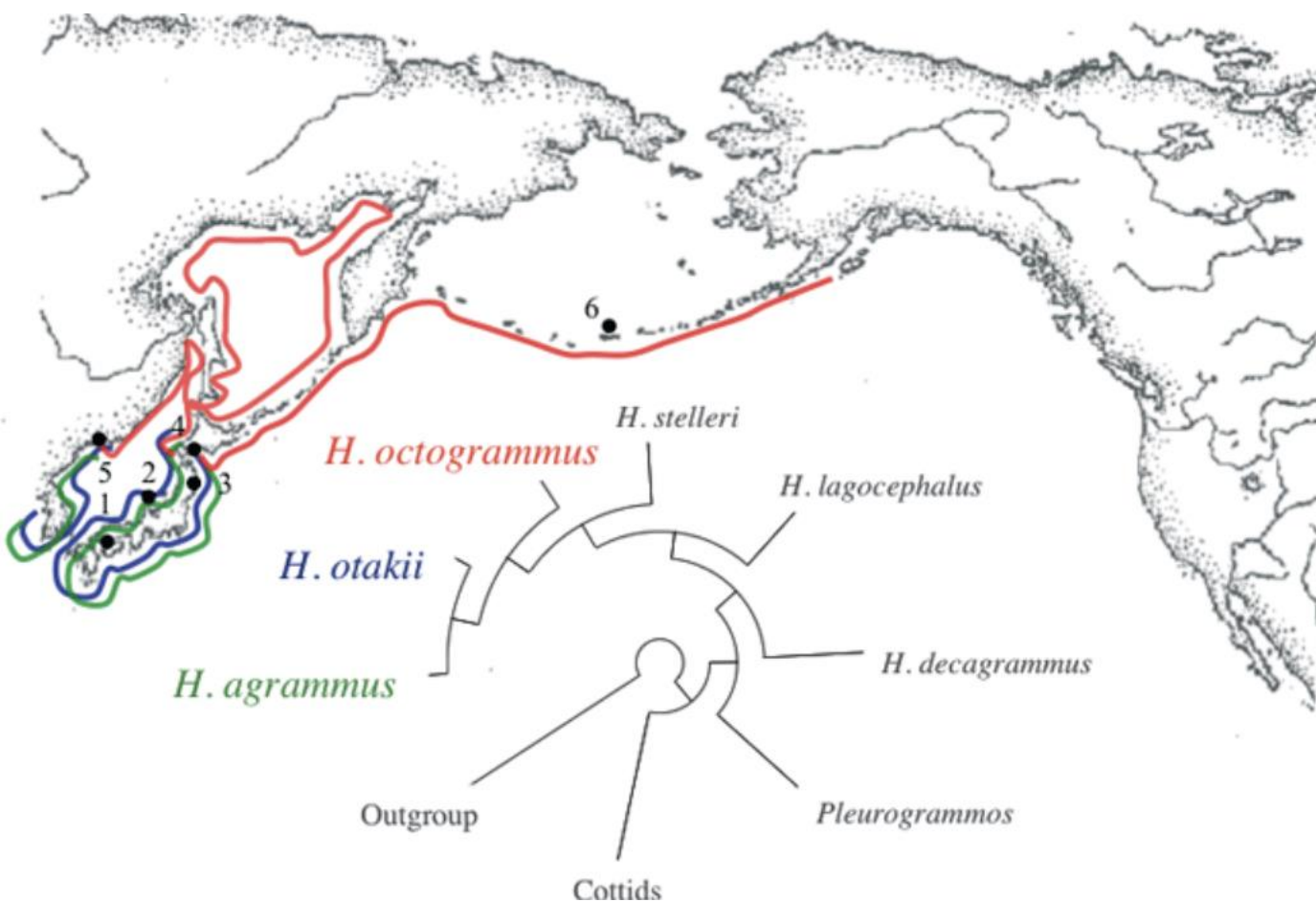


A photograph of a Fat greenling (Hexagrammos otakii) resting on a rocky seabed. The fish has a yellowish body with dark brown spots and stripes. It is positioned in front of several bright red, tubular sea anemones. The text "Fat greenling, Hexagrammos otakii" is overlaid in white at the top left.

Fat greenling,
Hexagrammos otakii

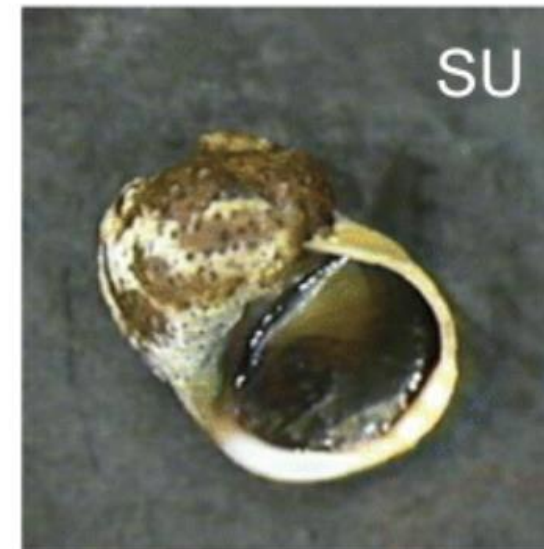
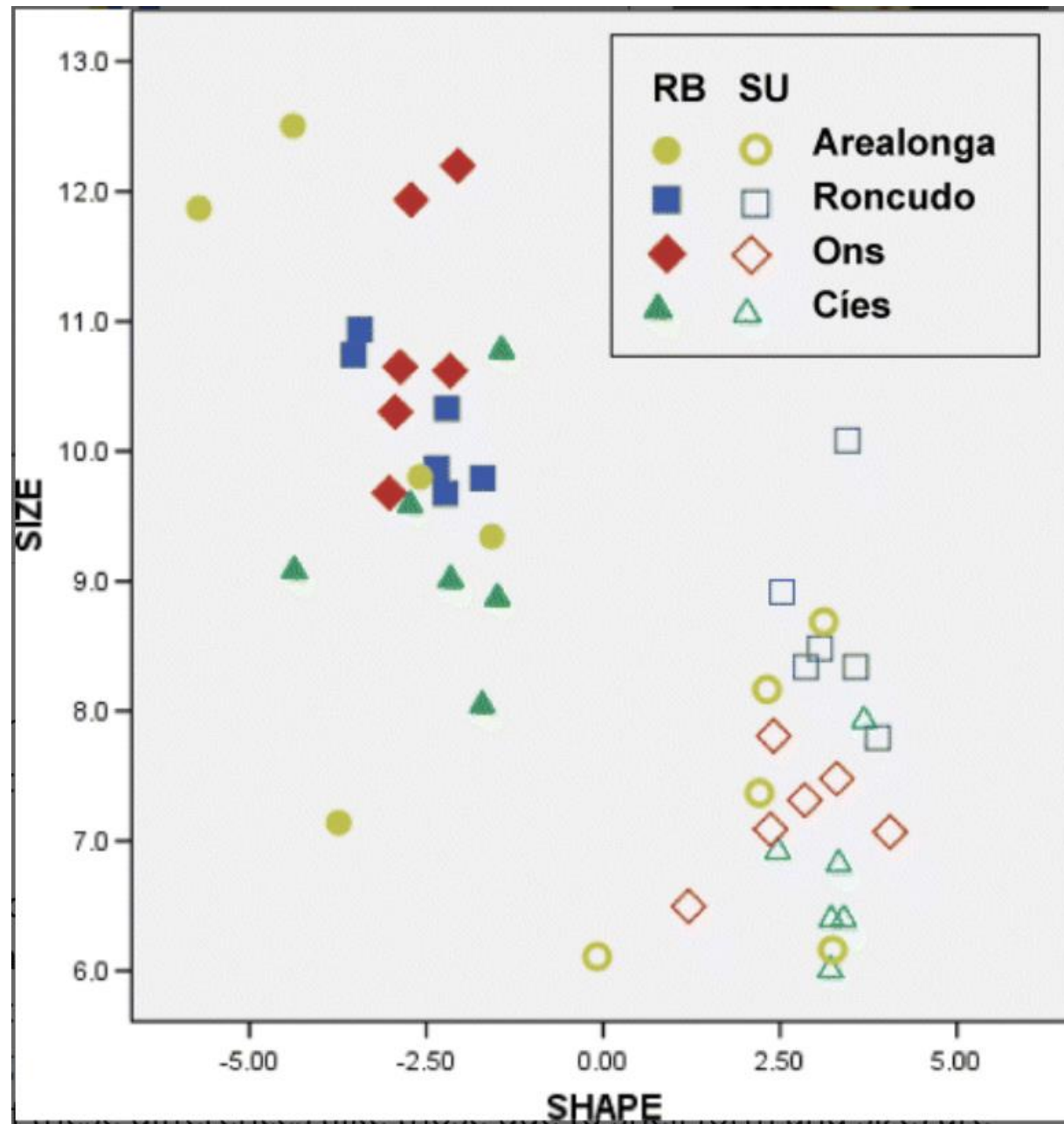
A close-up photograph of a Spotty-bellied greenling (Hexagrammos agrammus). The fish has a light-colored body with brown spots and a prominent yellow ring around its eye. It is shown in profile, swimming over a dark, rocky seabed. The text "Spotty-bellied greenling, Hexagrammos agrammus" is overlaid in white at the bottom right.

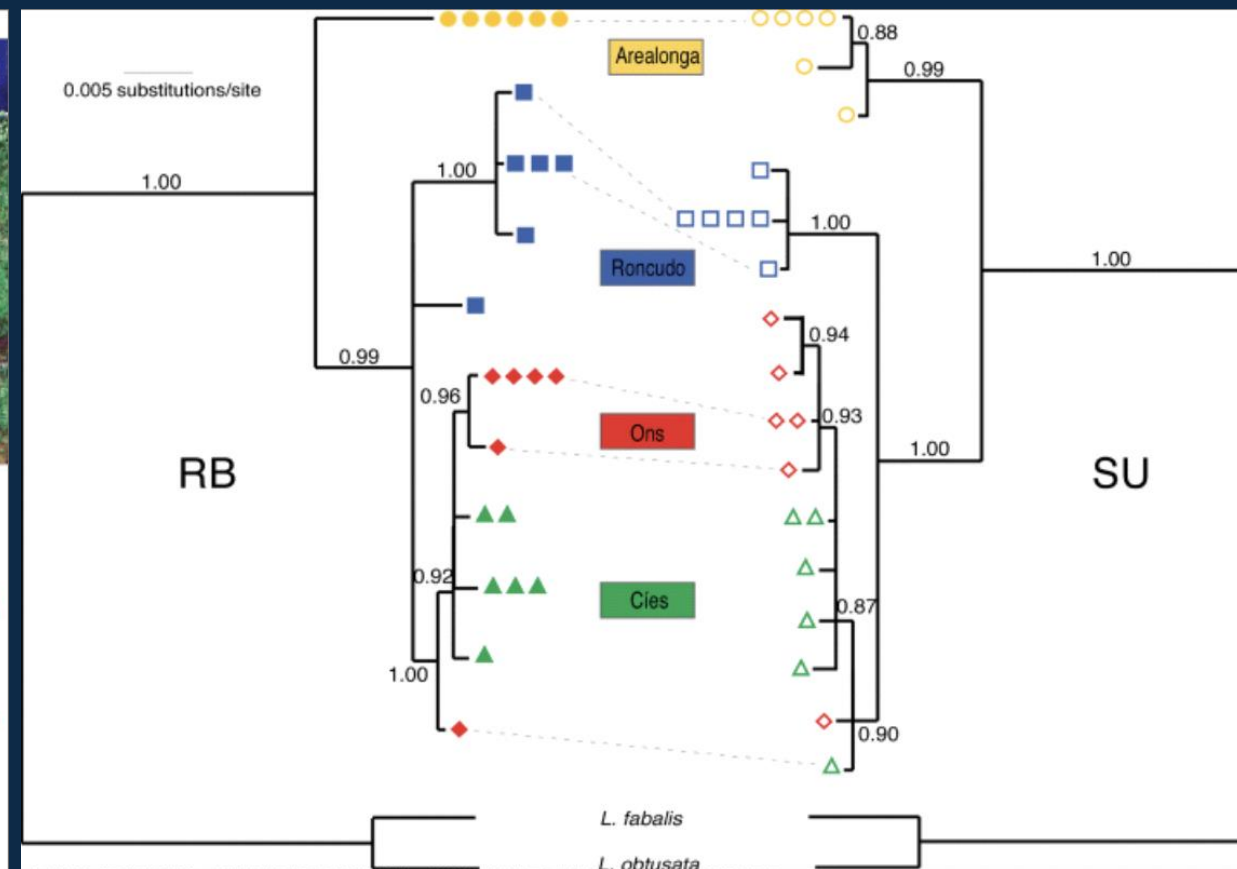
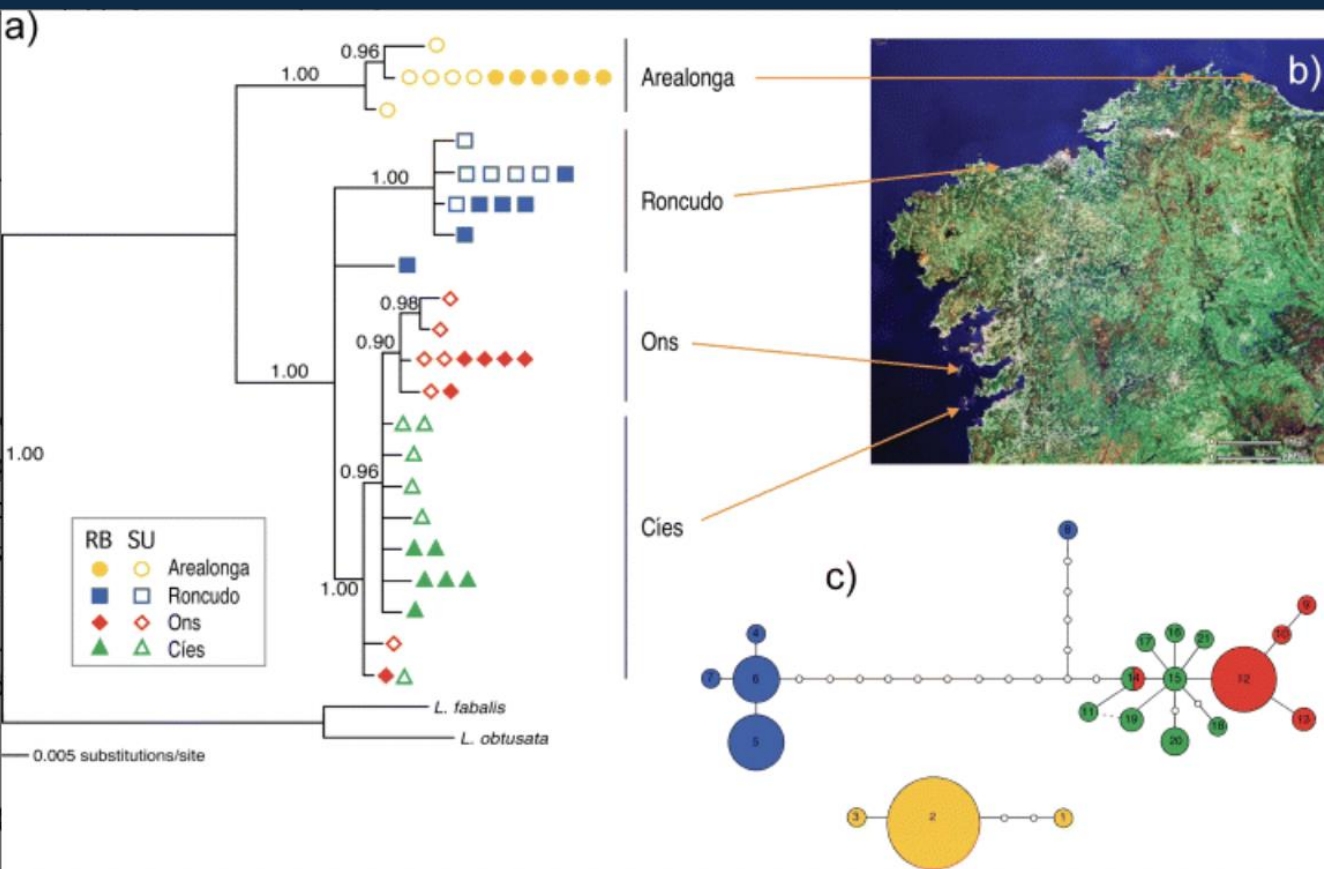
Spotty-bellied greenling,
Hexagrammos agrammus



	<i>H. octogrammus</i>	<i>H. agrammus</i>	<i>H. otakii</i>	Source
Reproduction				
Spawning season	October	September–October	Late October–November	Figure S1 and Sekigawa (2003)
Depth distribution of nests (m)	0.5–3.0	0.5–3.5	6.5–19.0	Kimura & Munehara (2010)
Nest substrate	Algal turf	Algal turf	Rock outcrops	Kimura & Munehara (2010)
Morphology				
Average SL (cm)	18.1 ($\pm$, $n = 38$)	20.5 (+, $n = 97$)	28.2 (+, $n = 42$)	Crow <i>et al.</i> (2007)
Lateral line number	5	1	5	Illustrated in Crow <i>et al.</i> (2007)
Caudal fin	Round	Round	Square	Illustrated in Crow <i>et al.</i> (2007)
Nuptial colouration	Pelvic anal fin black	Pelvic anal fin black	Body bright yellow	Munehara <i>et al.</i> (2000)



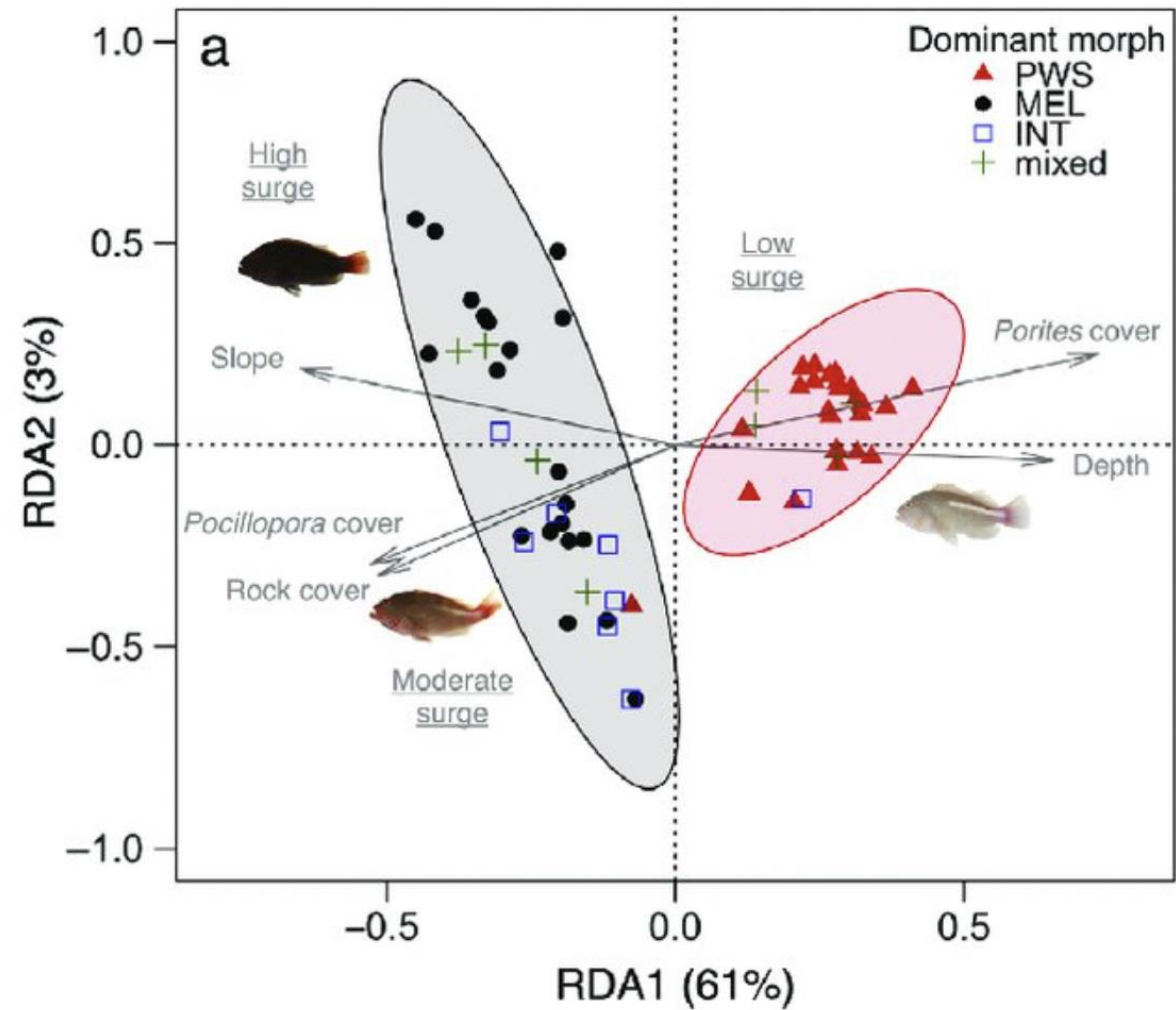
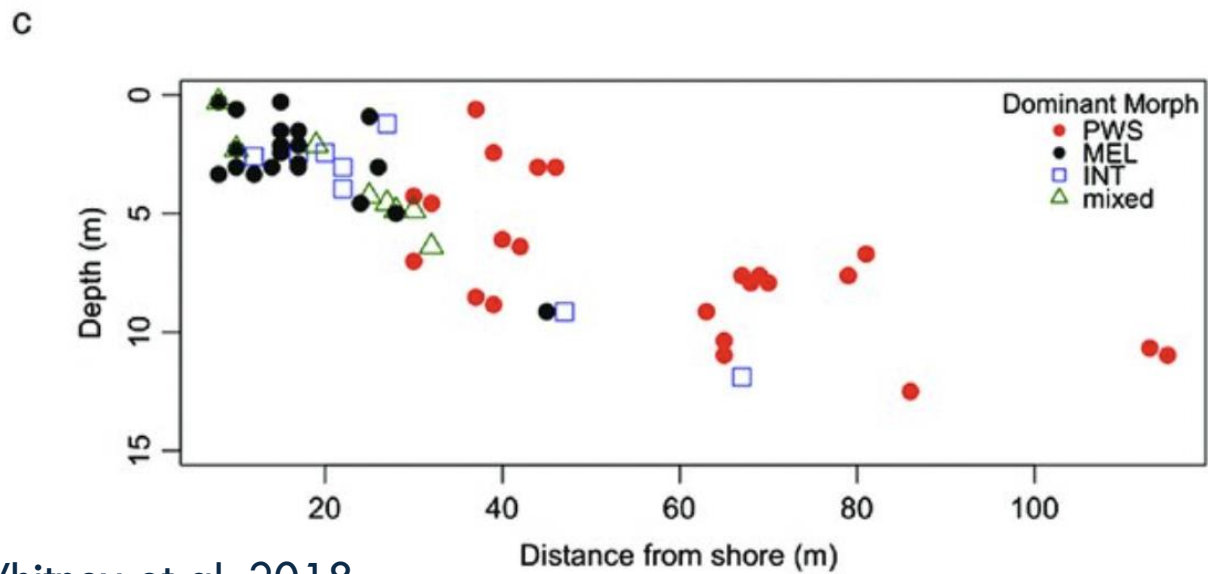
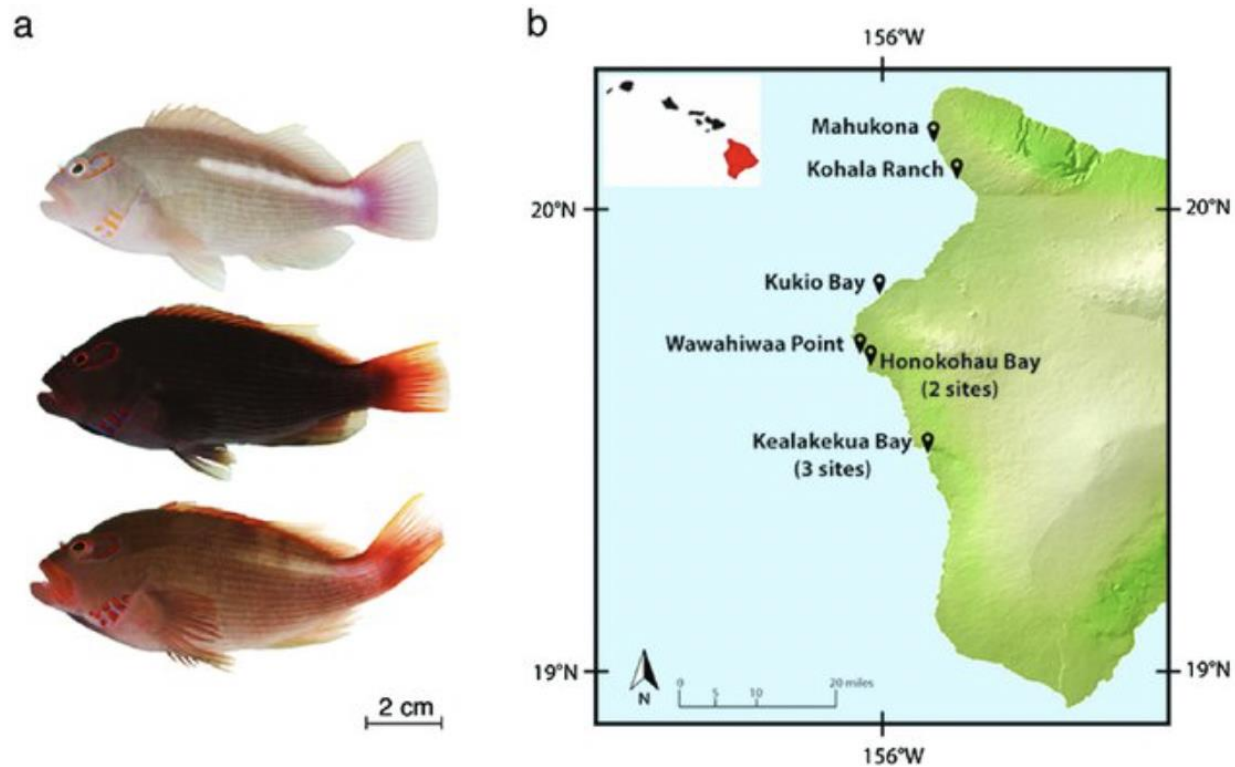






33 clear cases of ecological speciation:

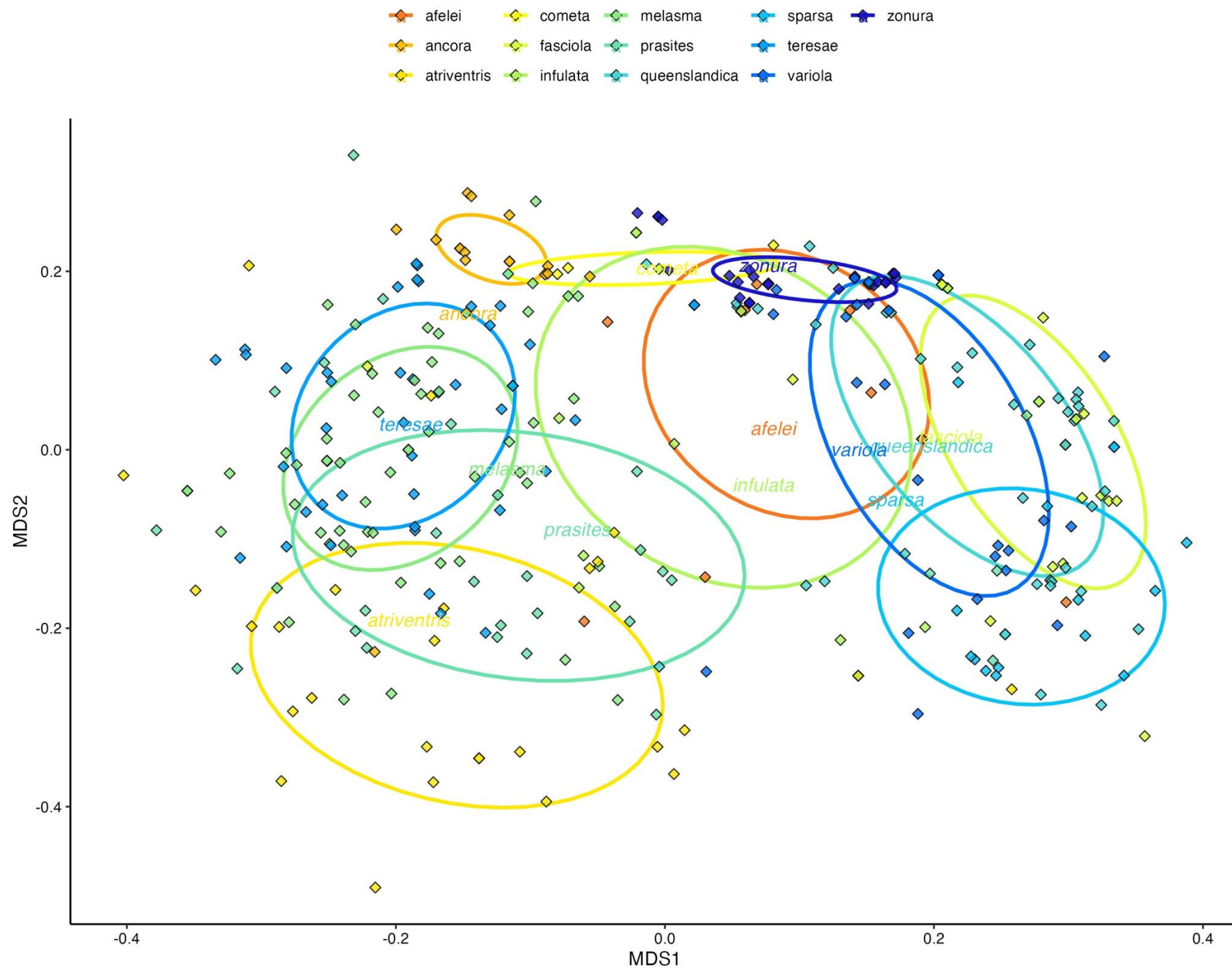
- 21 tropical, 12 temperate/polar
- 11 fishes, 6 gastropods, 2 shrimps, 3 corals

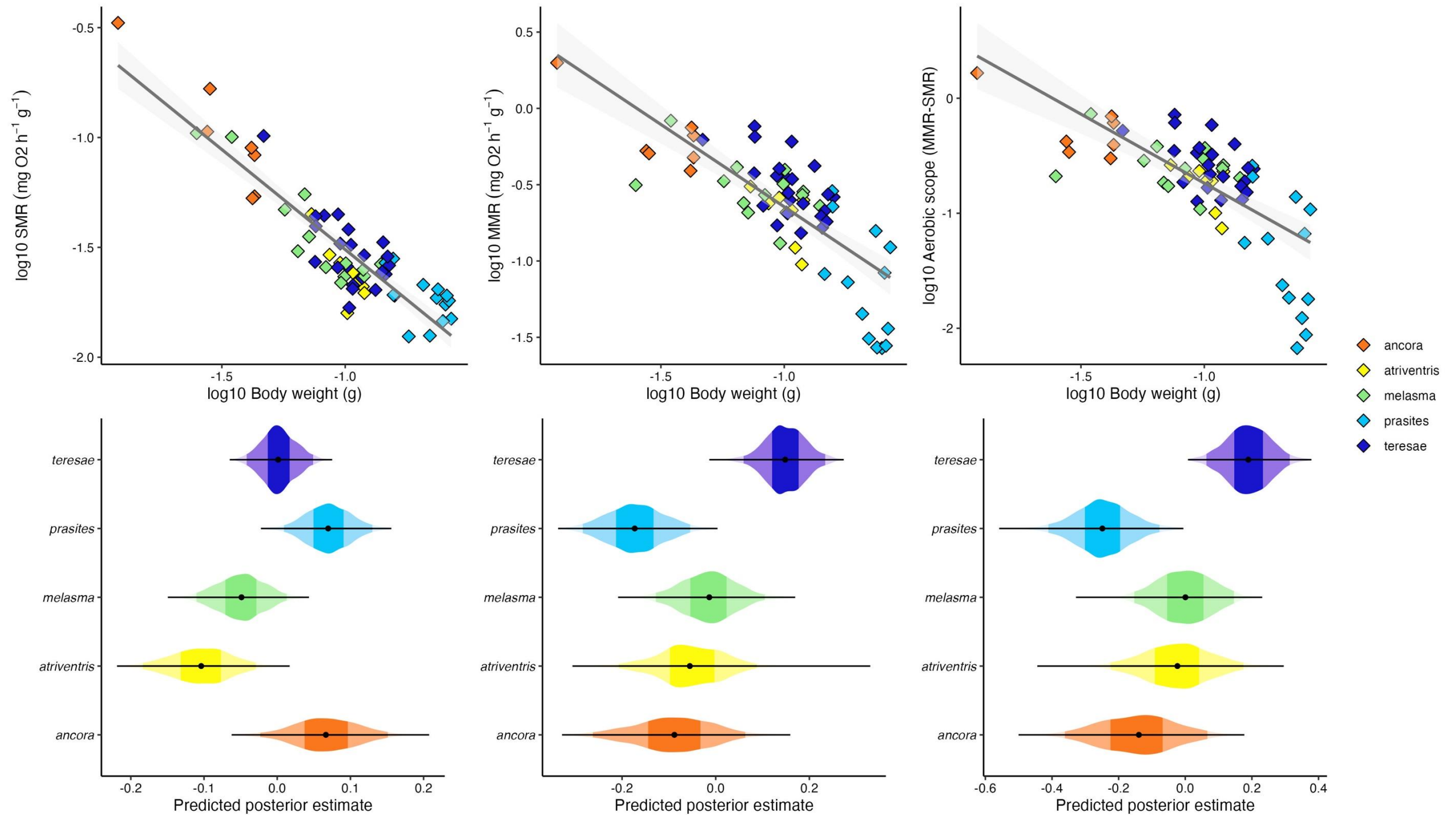


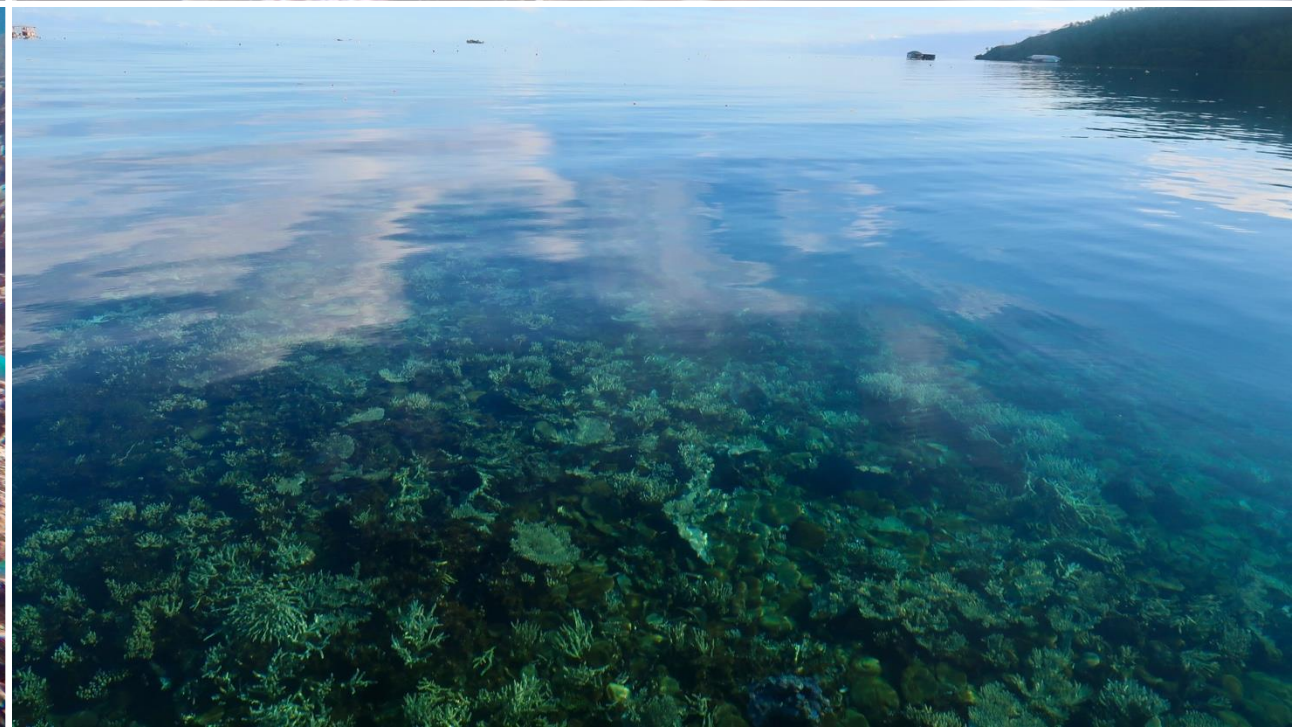
1. Are populations reproductively isolated?
2. Is there (potential for) disruptive selection?
3. Do populations mate assortatively in sympatry?
4. Is the selected trait linked to the assortment trait?
5. Is there evidence for gene flow between populations at the time of divergence?
6. Do geographic ranges of populations overlap?

Ecological speciation is faster
than allopatric speciation

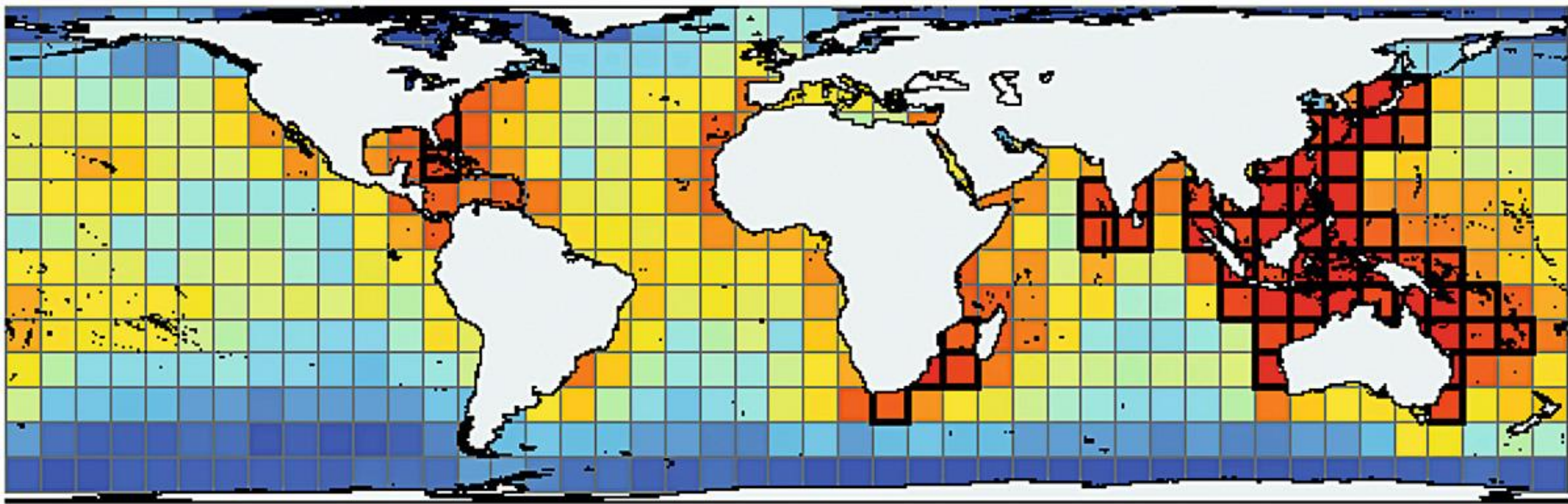








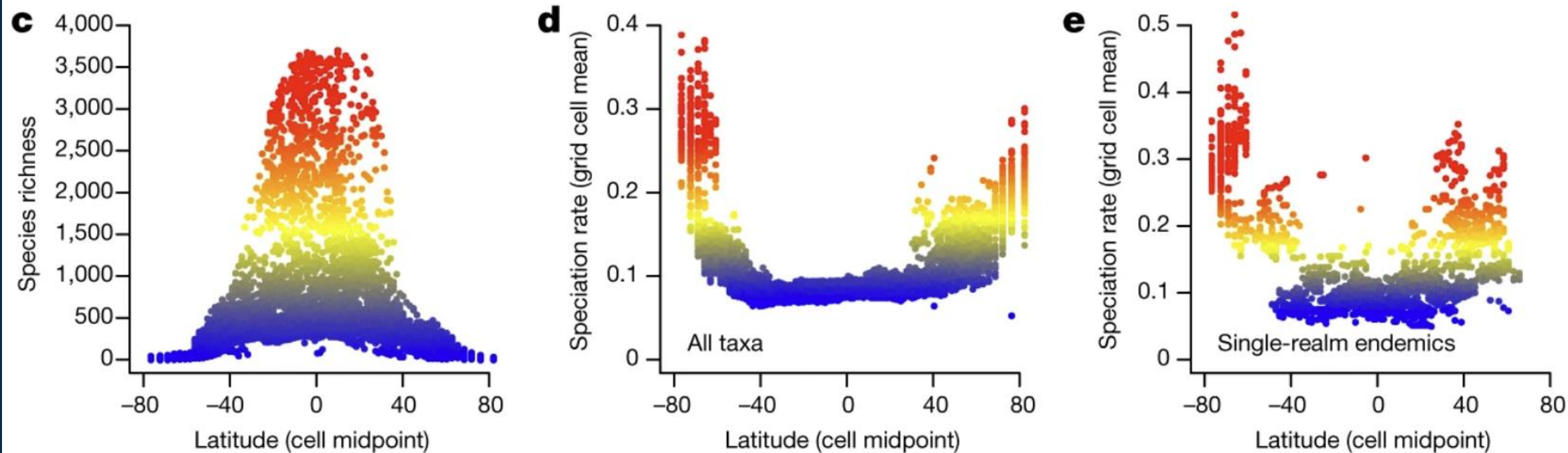
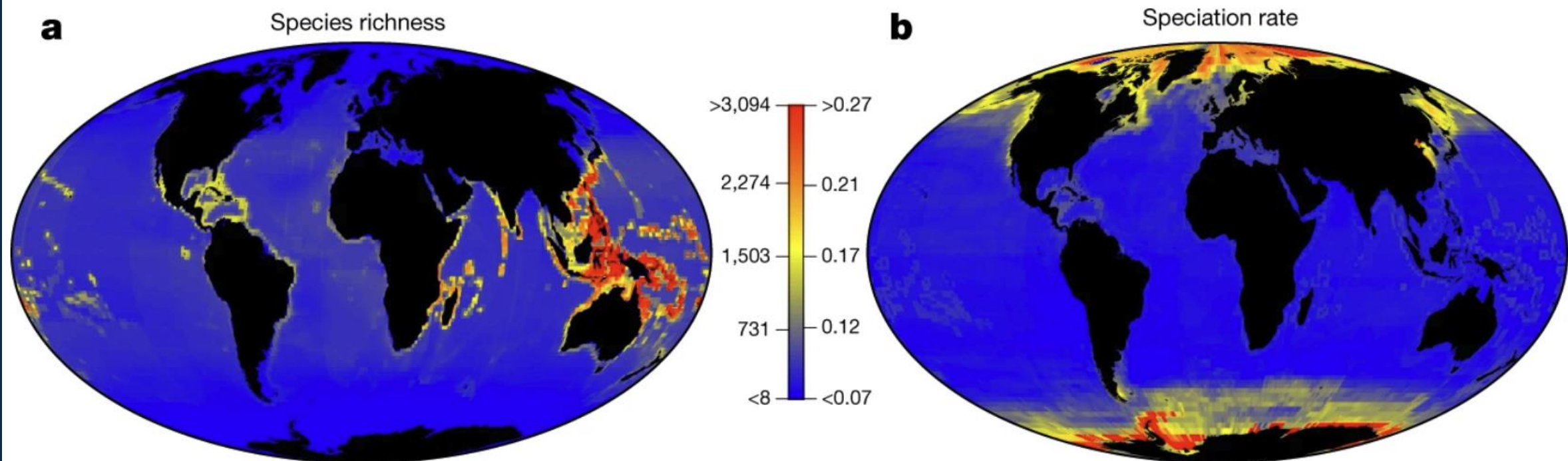
Global biodiversity patterns

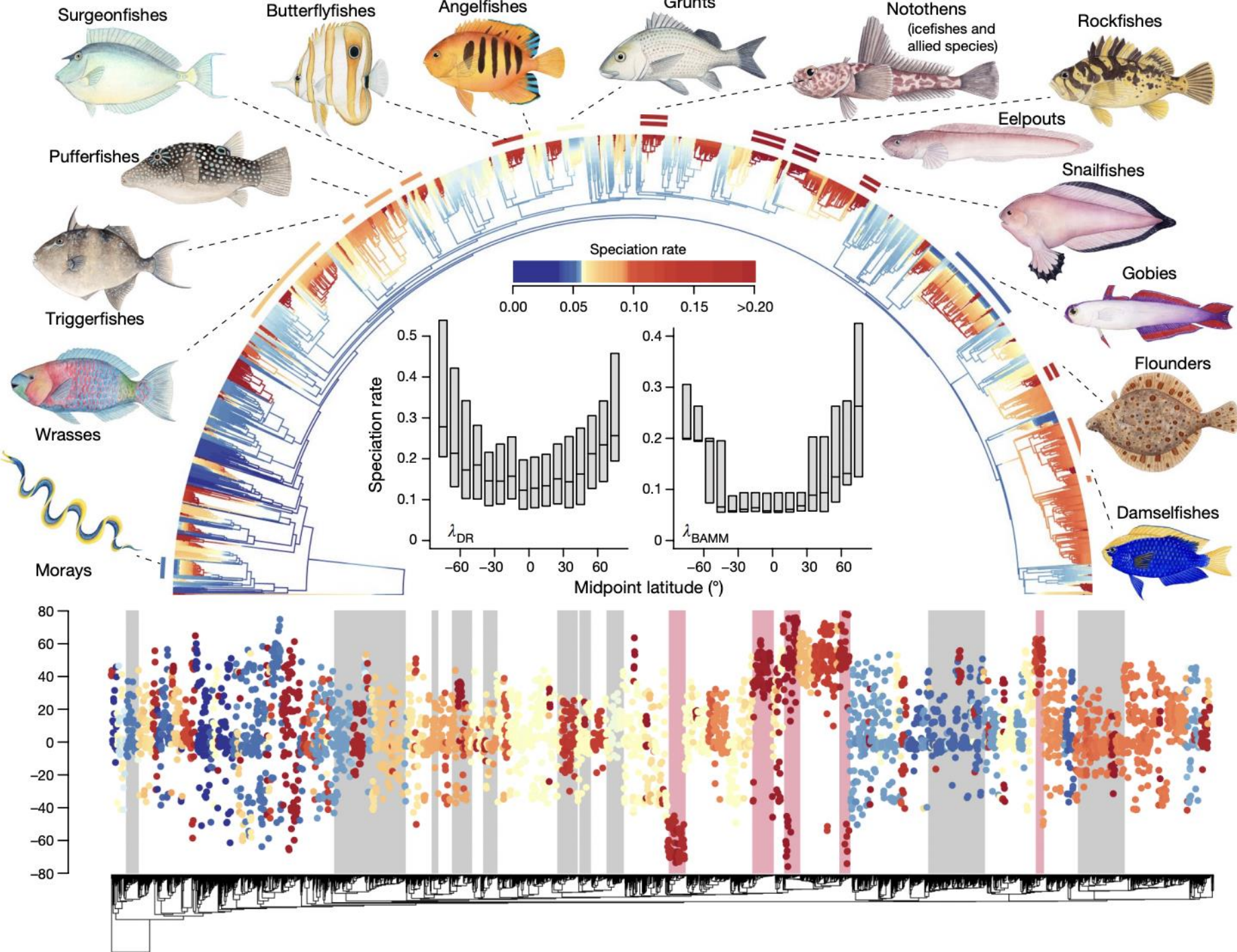


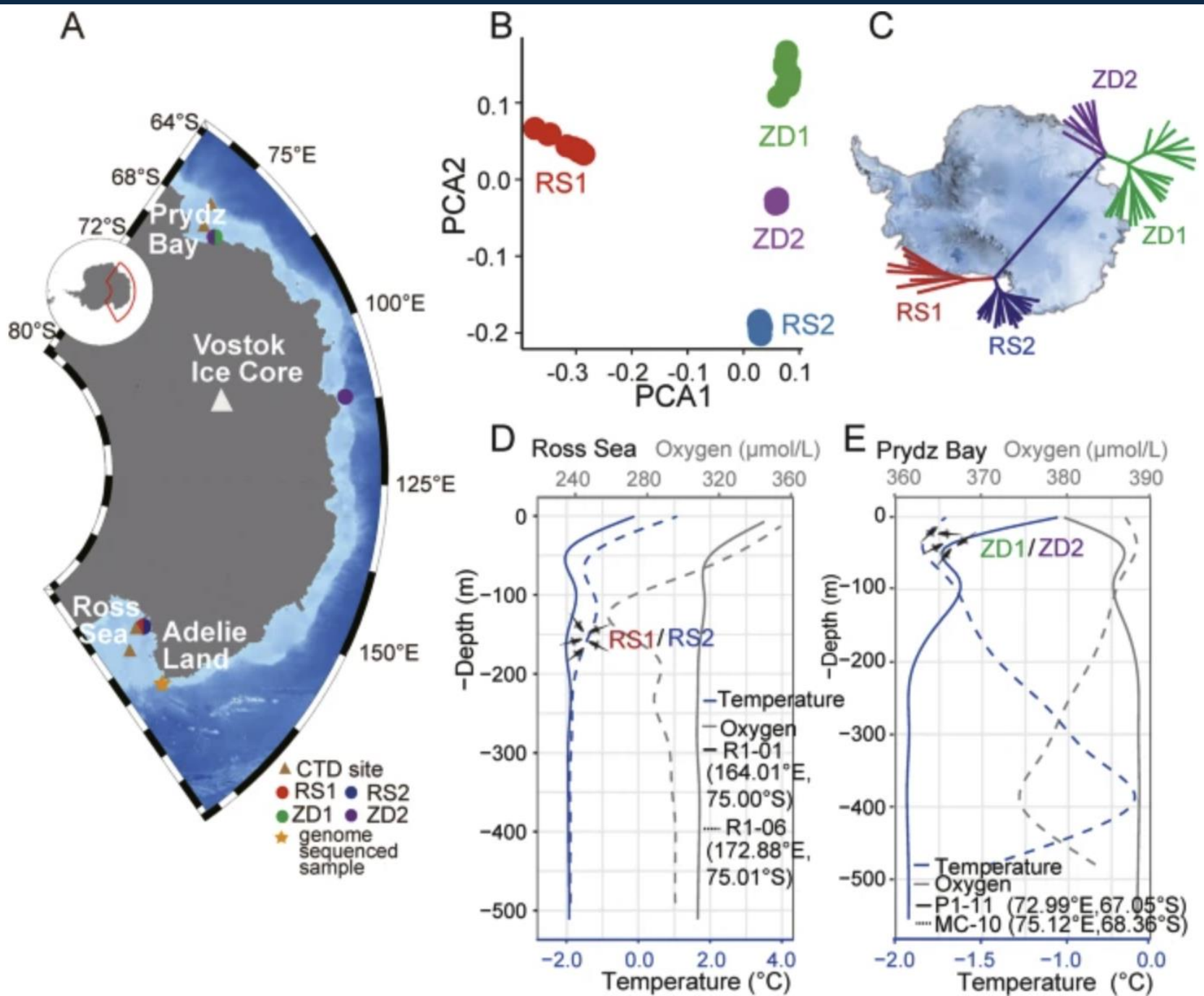
Channichthyidae: notothenioid icefishes





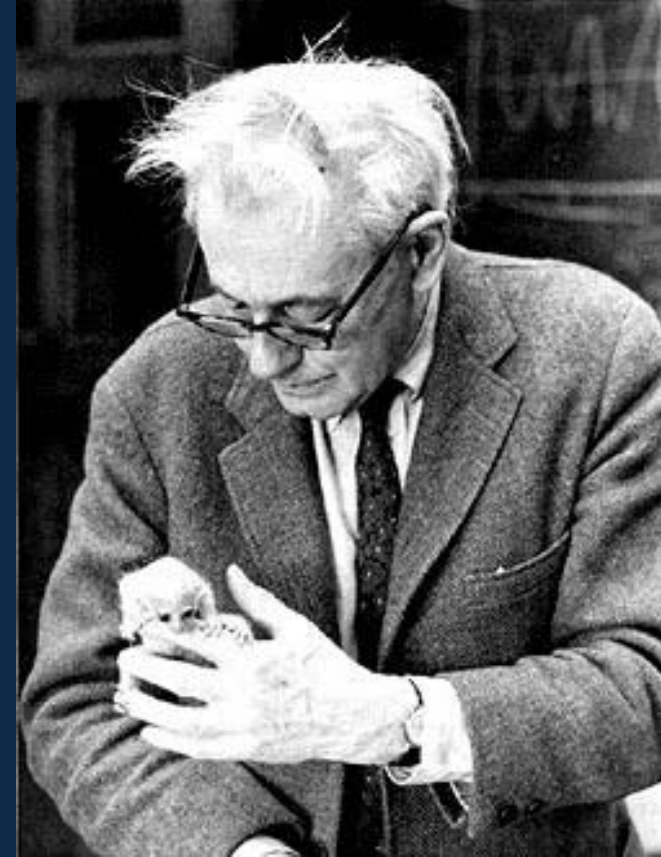






It is certain that the fauna of any such region is qualitatively poorer than that of warm temperate and tropical areas of comparable effective precipitation. It is probably considered to be intuitively obvious that this should be so, but on analysis the obviousness tends to disappear. If we can have one or two species of a large family adapted to the rigors of Arctic existence, why can we not have more?

Hutchinson (1959)



Discussion

Take 5 minutes to discuss:

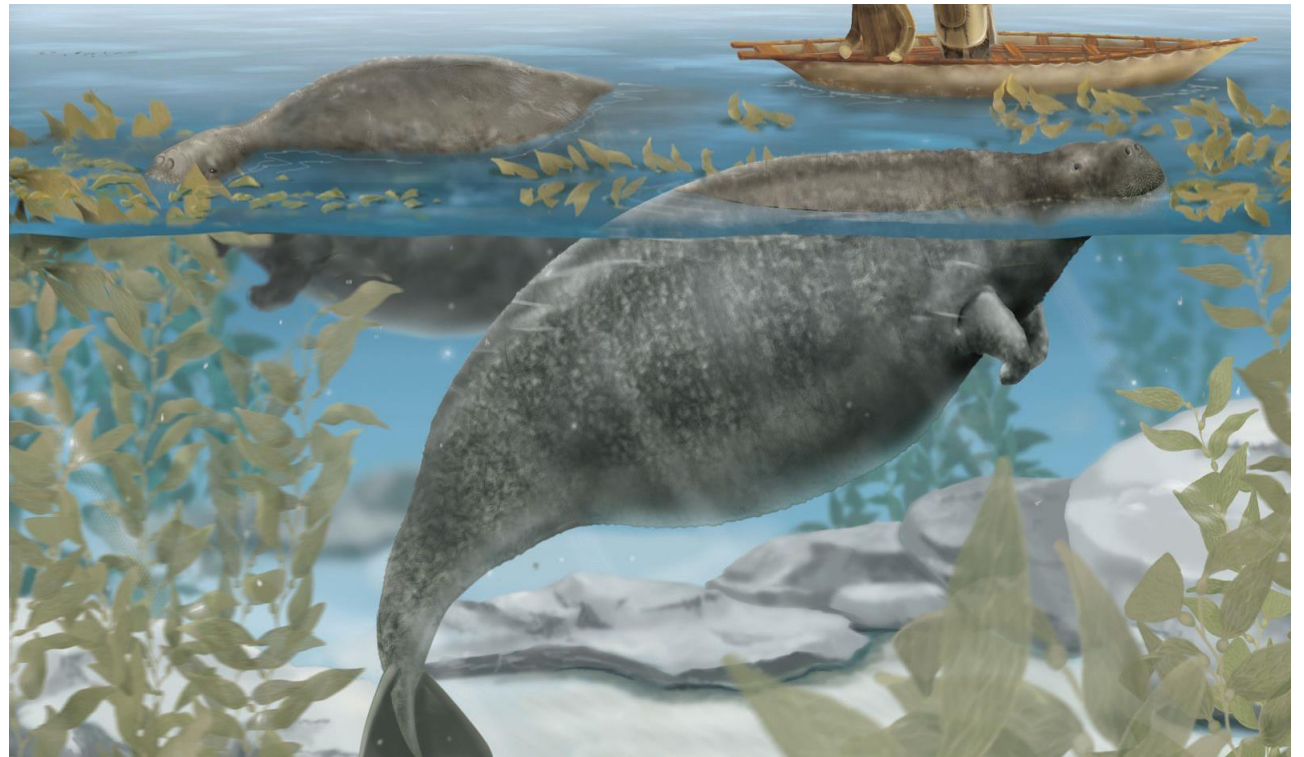
- 1) Why can't we have more?



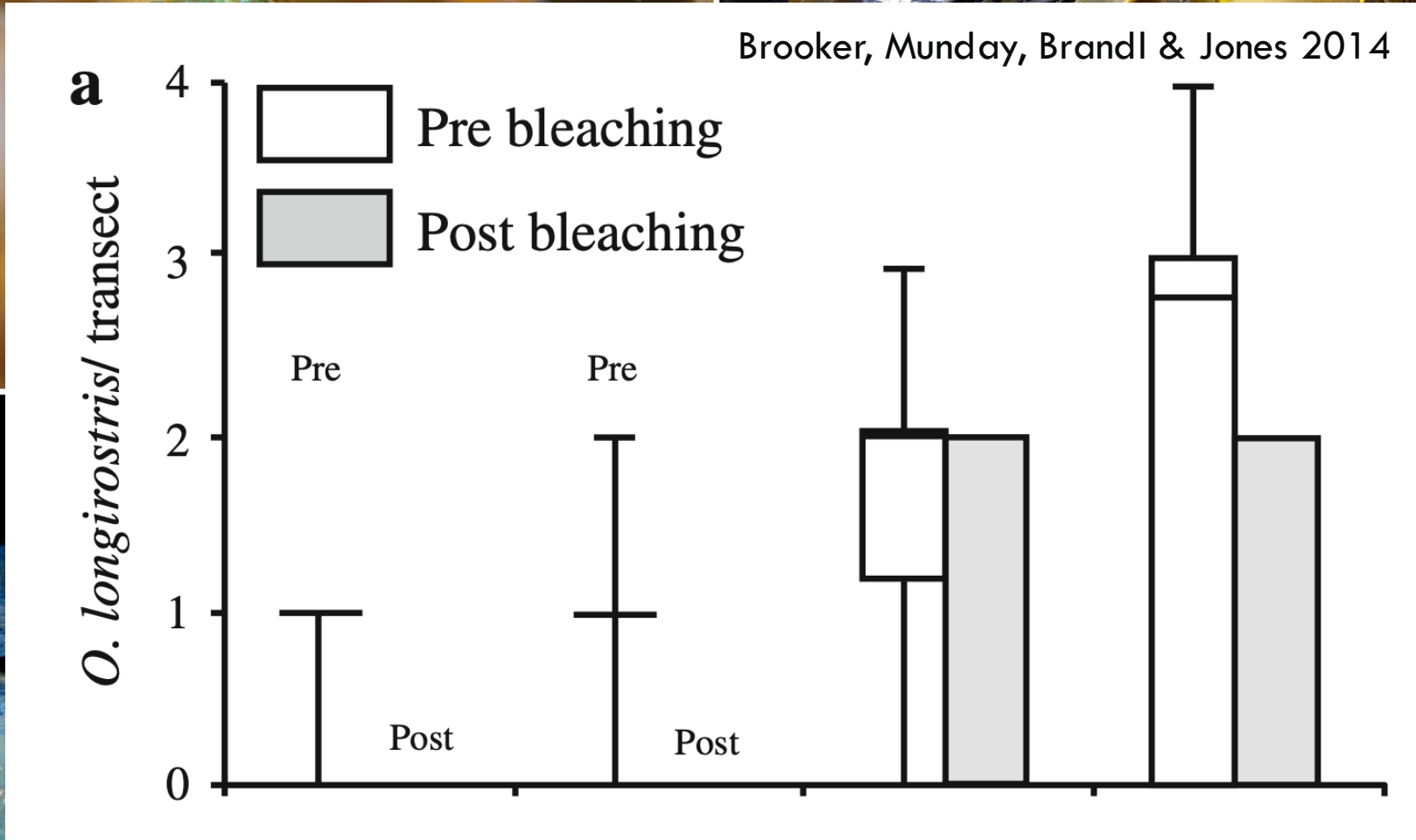
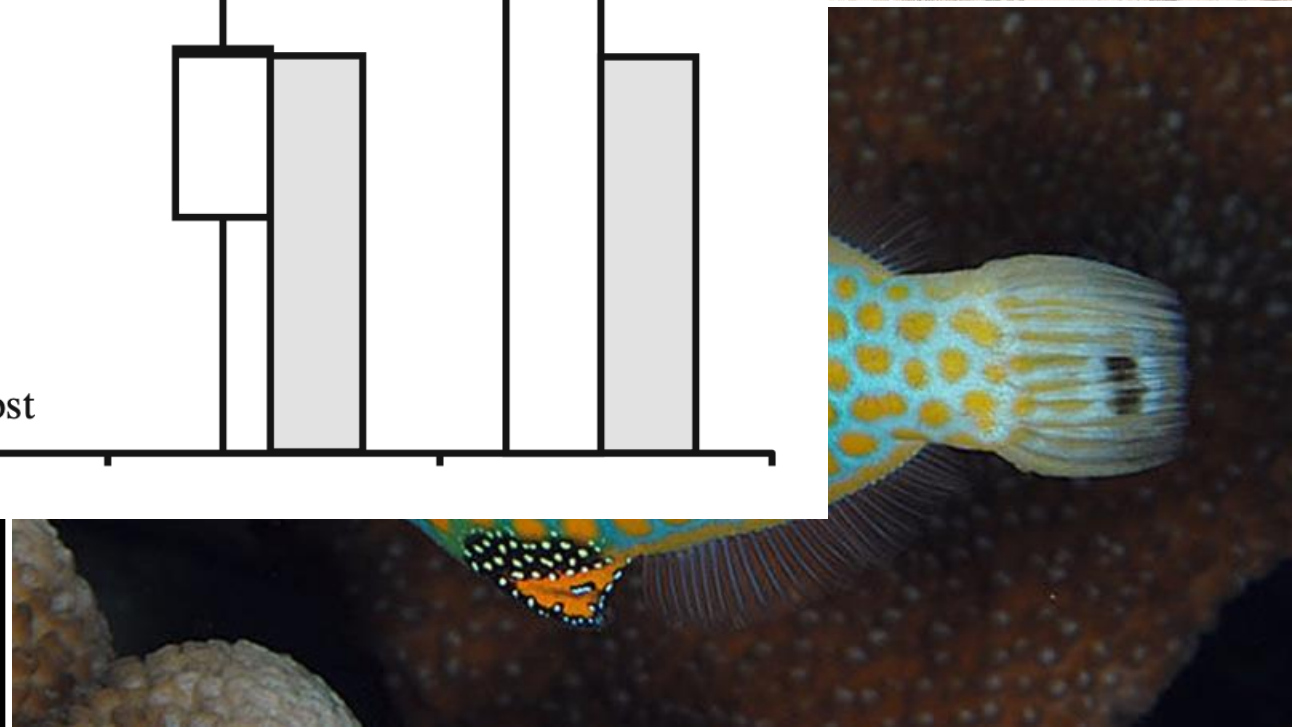
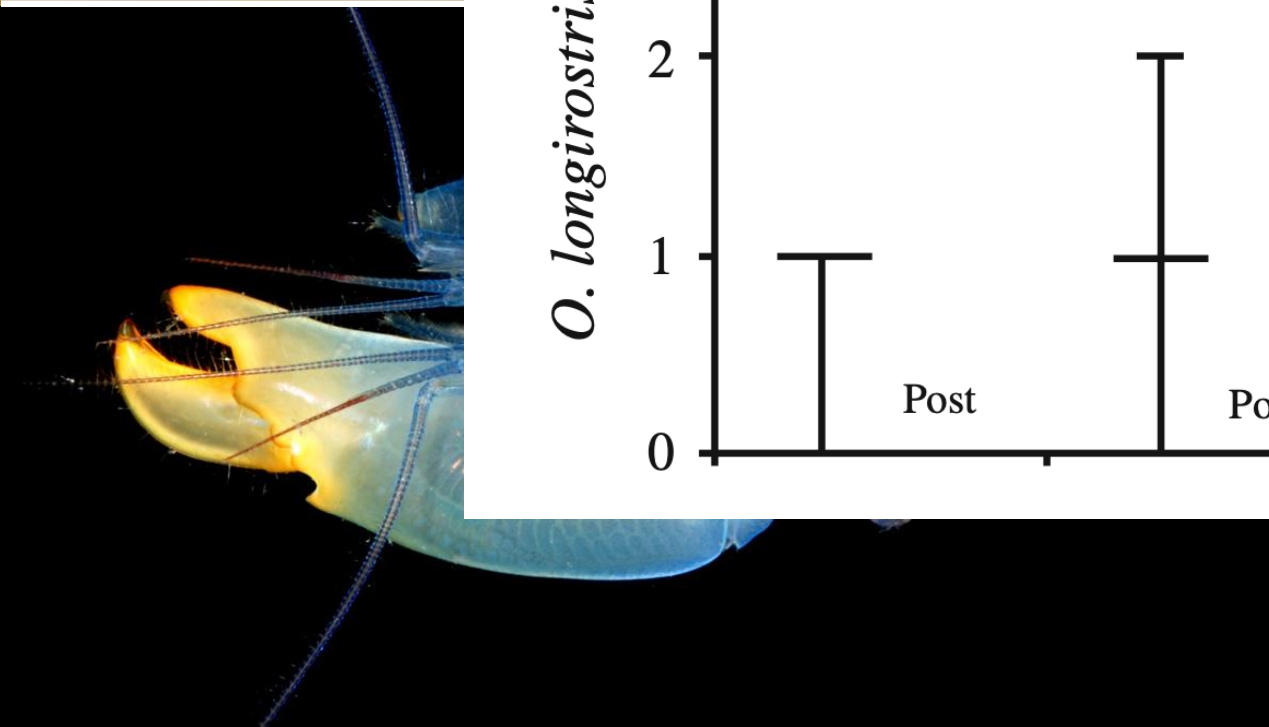
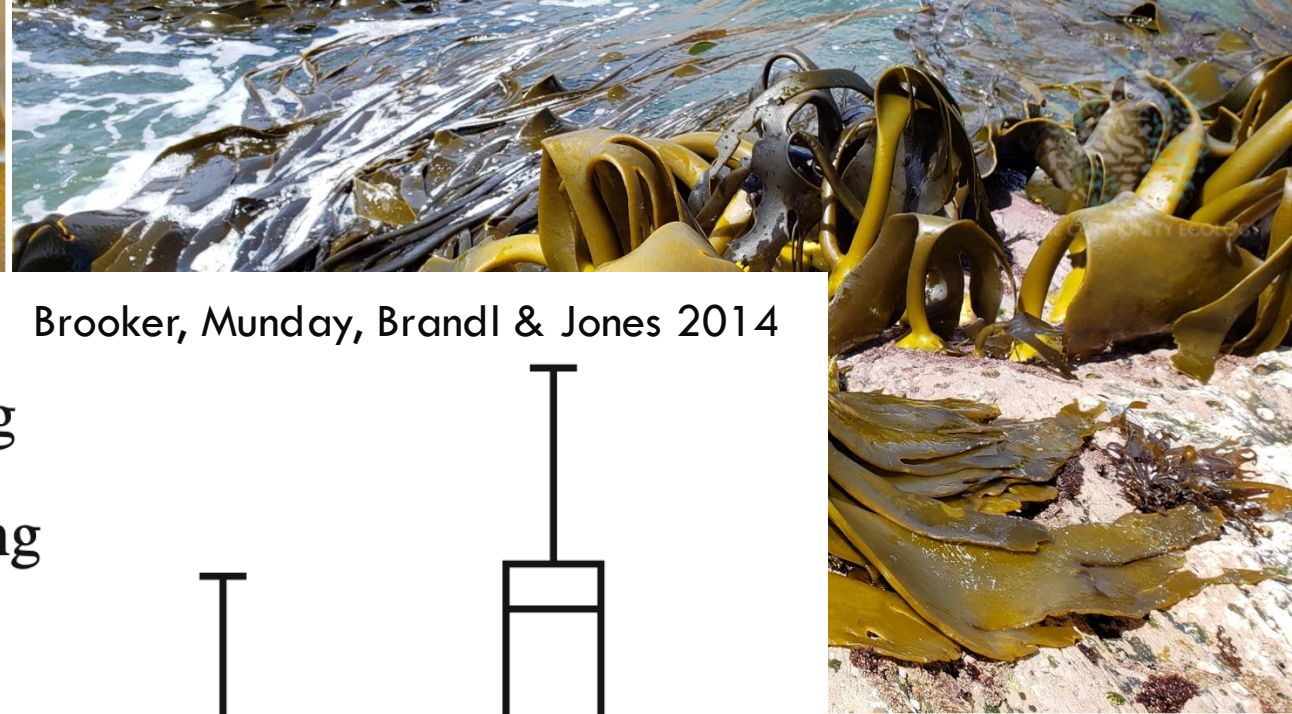
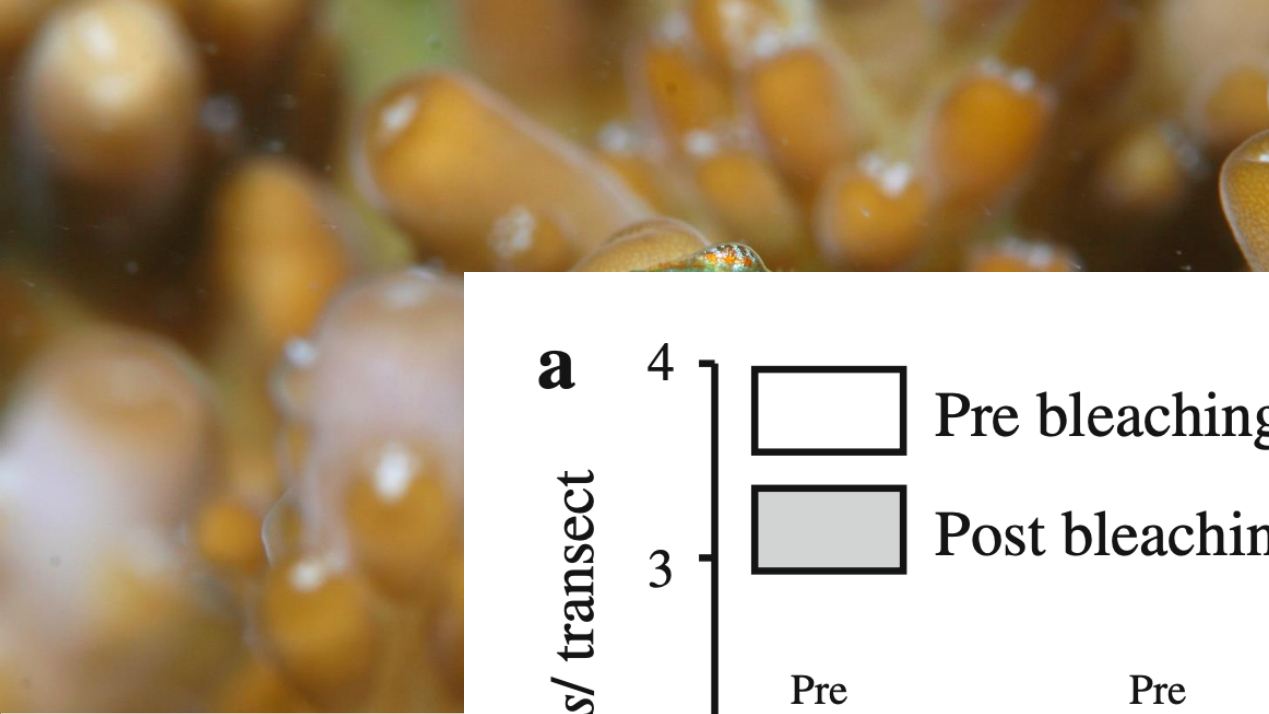


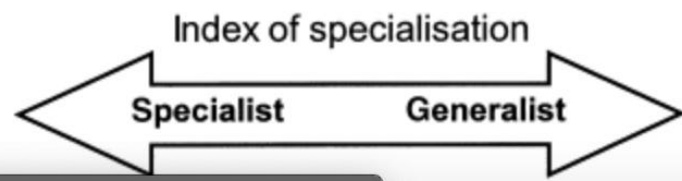
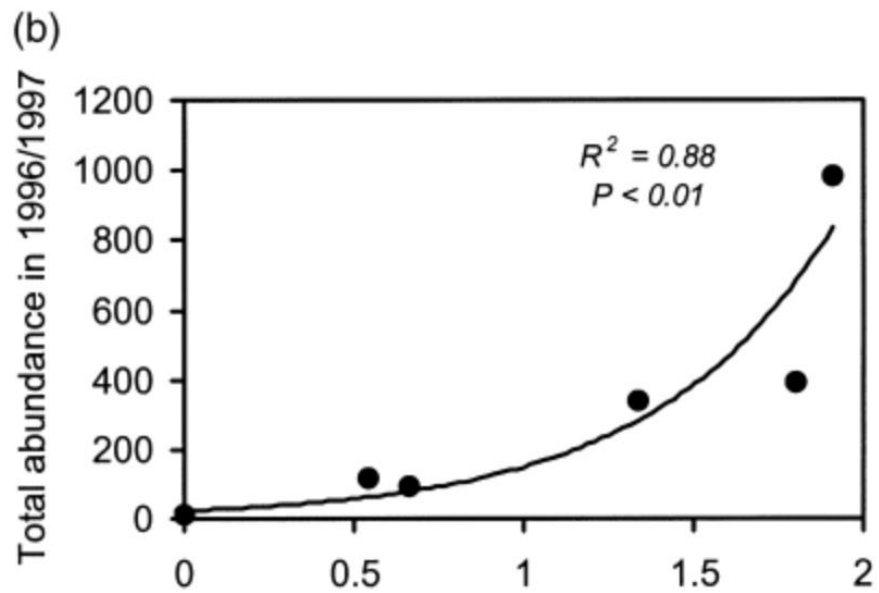
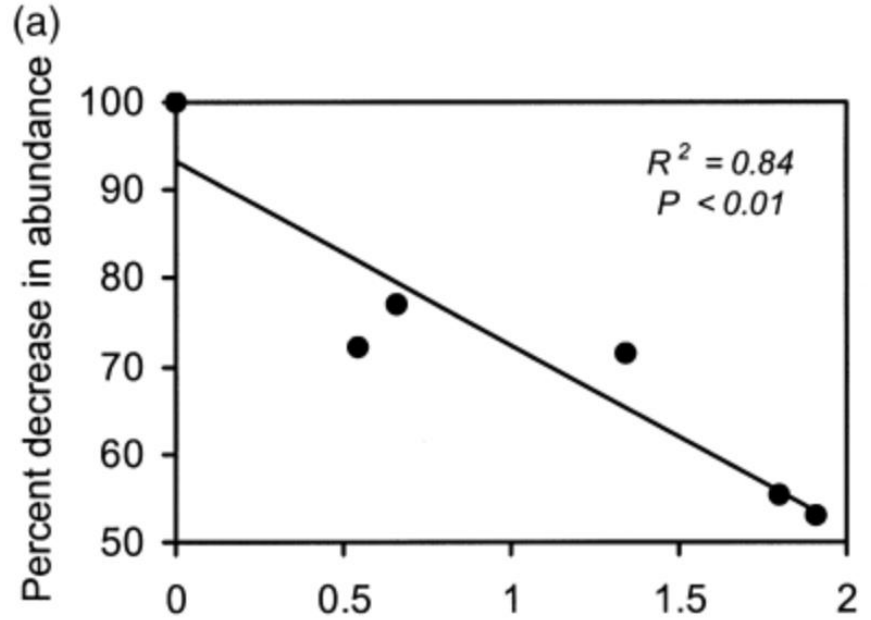


Drift



- 1) Global extinction
- 2) Local extinction
- 3) Ecological extinction





Gobiodon species A



Gobiodon species B



Members of this family possess modified pectoral fins resembling hands with which they use to preferentially “walk” across the sea floor instead of swimming. Handfishes exhibit various qualities that make them particularly vulnerable to extinction, such as lacking a dispersive pelagic larval stage, and high energetic cost to reproduction, investing in fewer, larger eggs that are guarded by the mother during development (Bruce et al., 1997; Last & Gledhill, 2009).



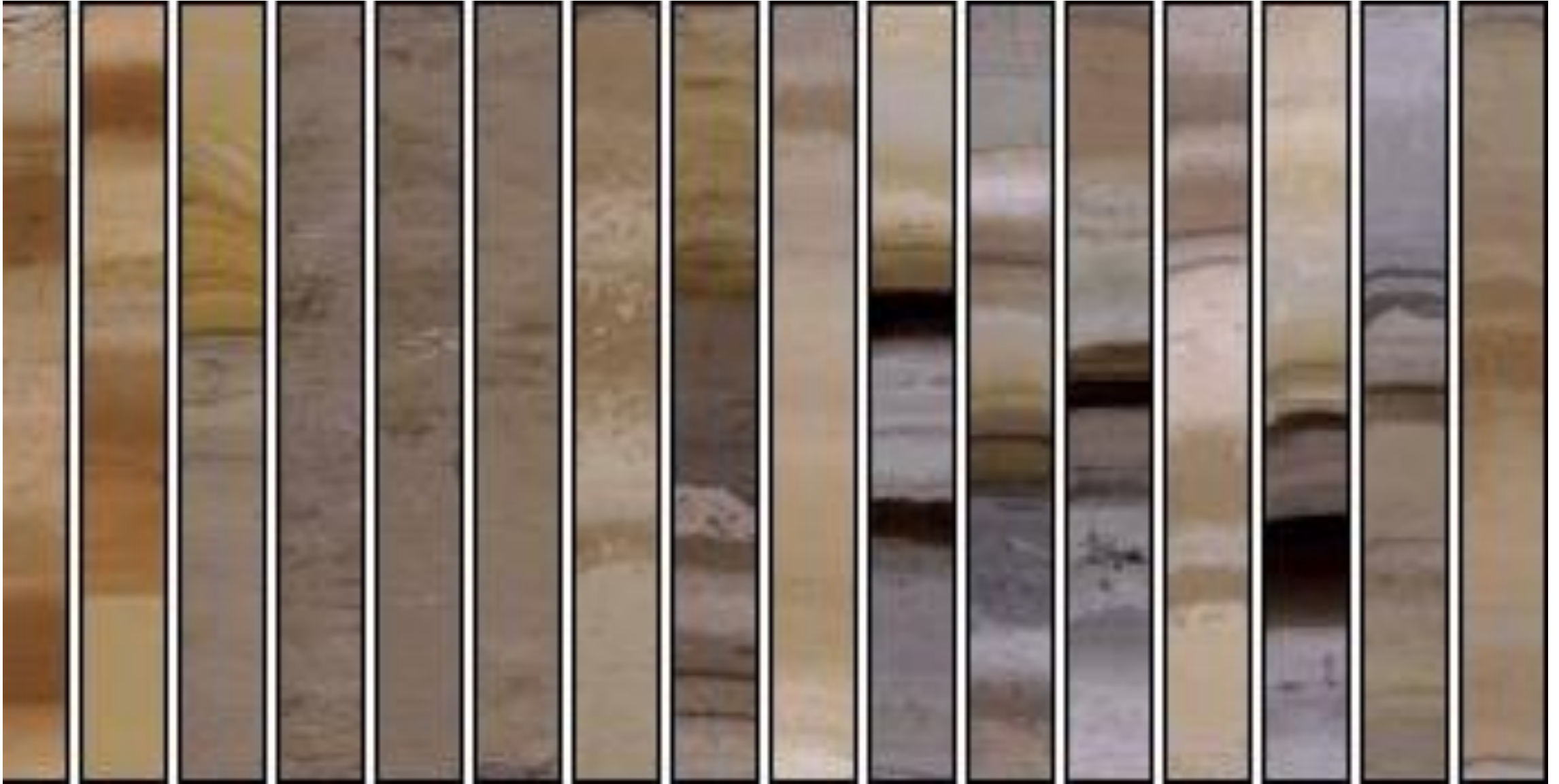
A close-up photograph of a red frogfish resting on a seabed covered in small, light-colored rocks and shells. The fish is a vibrant red color with a large, prominent dorsal fin and a smaller, spiky fin on its head. Its eyes are large and have a blue-green center. The text "101 individuals" is overlaid in white, sans-serif font across the middle of the fish's body.

101 individuals

**REMEMBER WHEN
THAT SPECIES WAS HERE?**



PEPPERIDGE FARM REMEMBERS



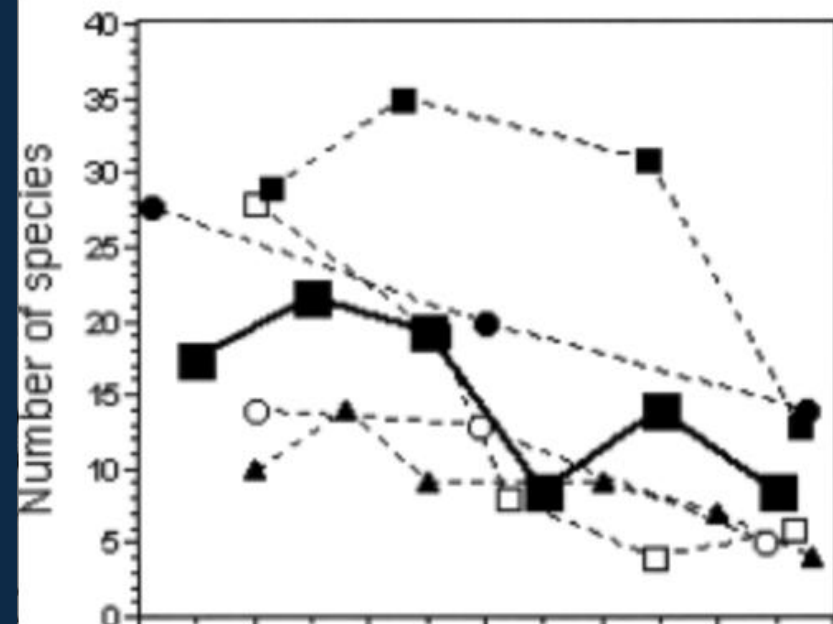


Po River Delta, Italy

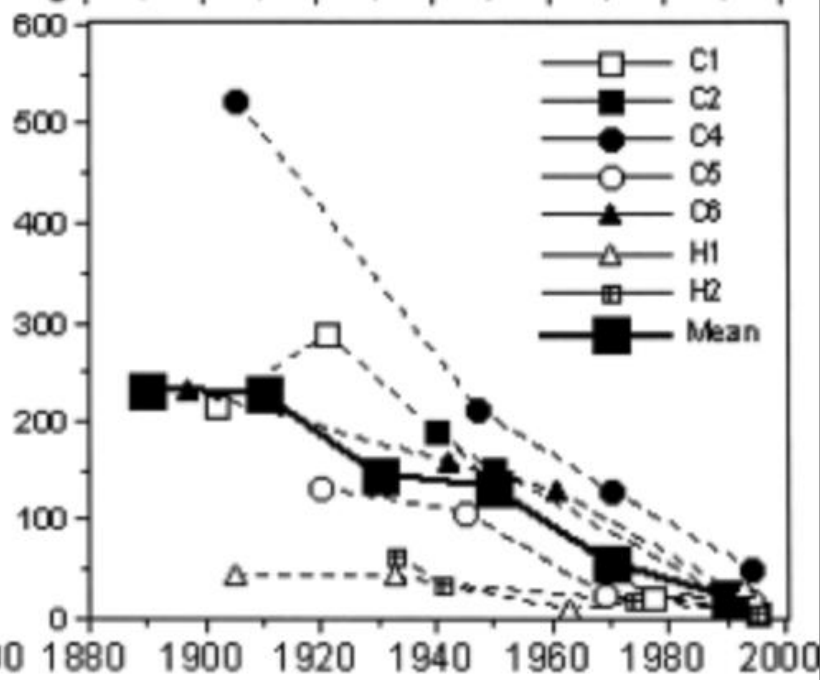
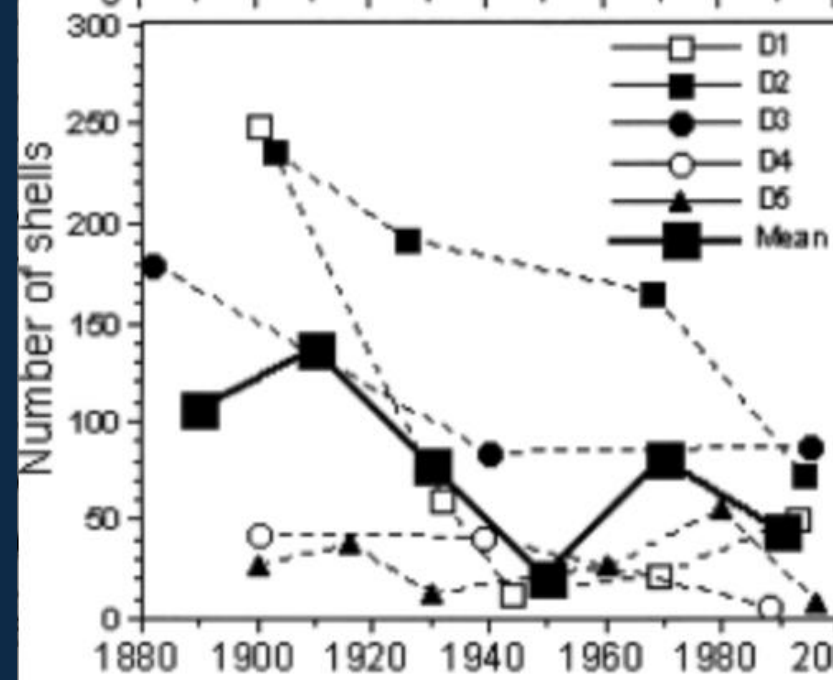
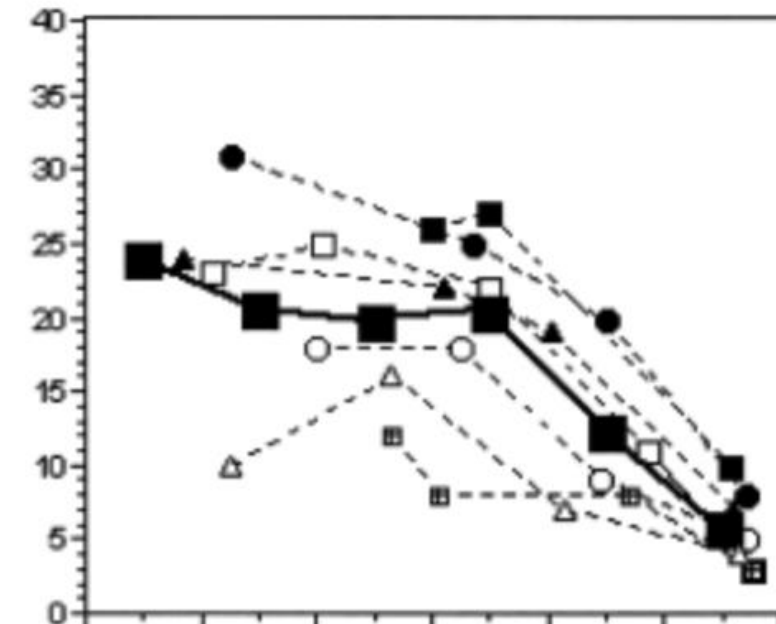
Tasmania, Australia



Derwent



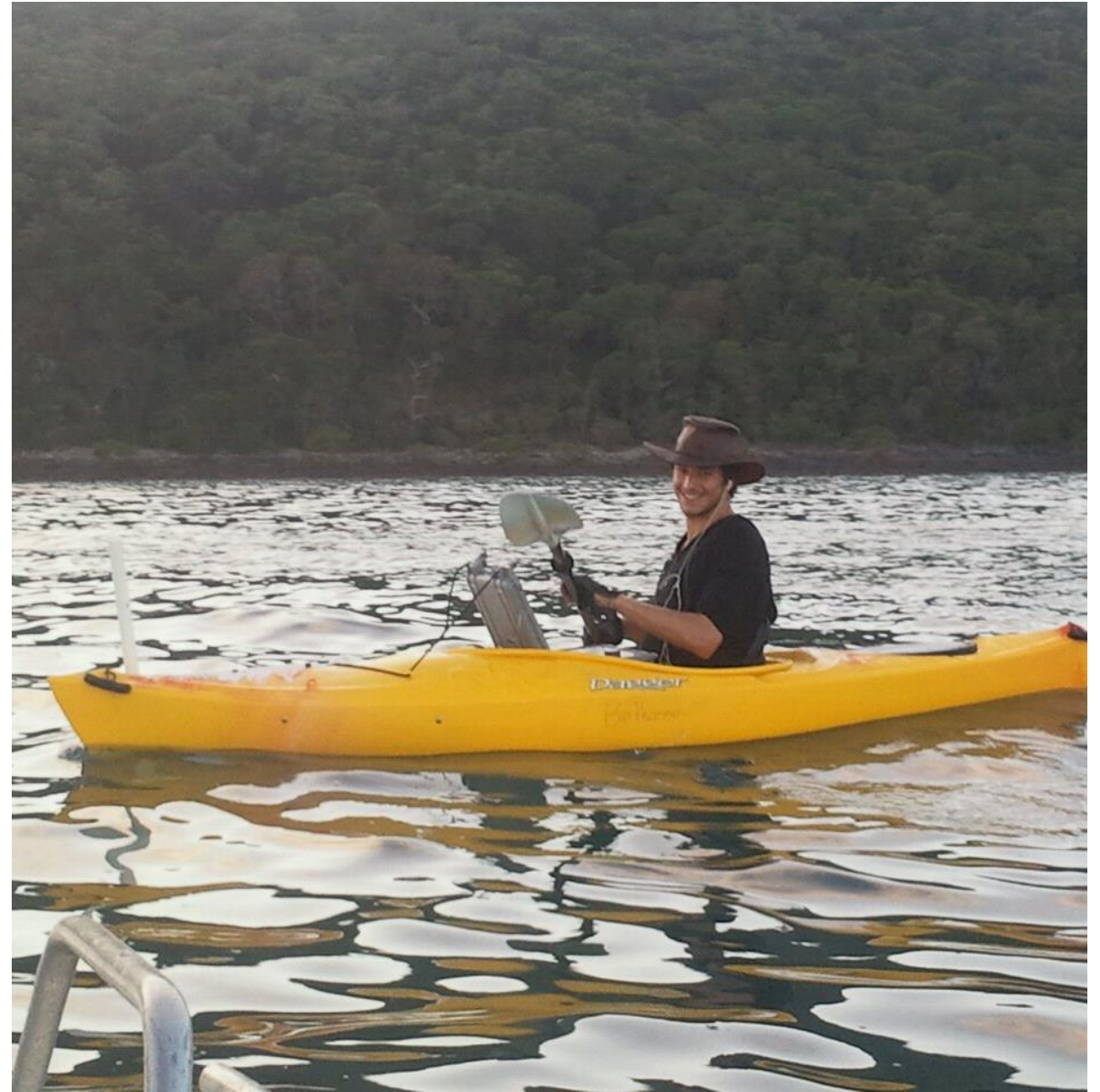
D'Entrecasteaux/Huon

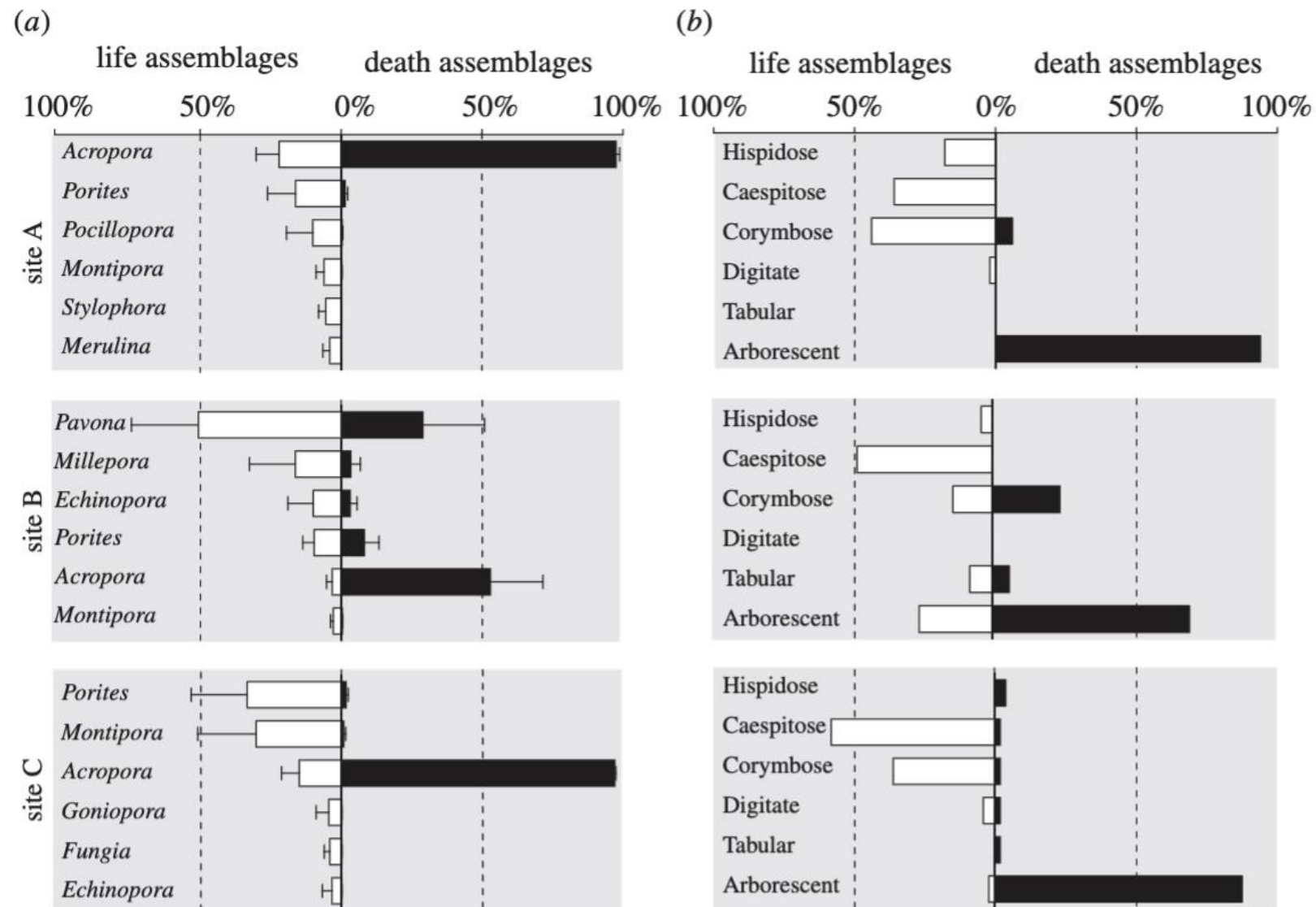


Year

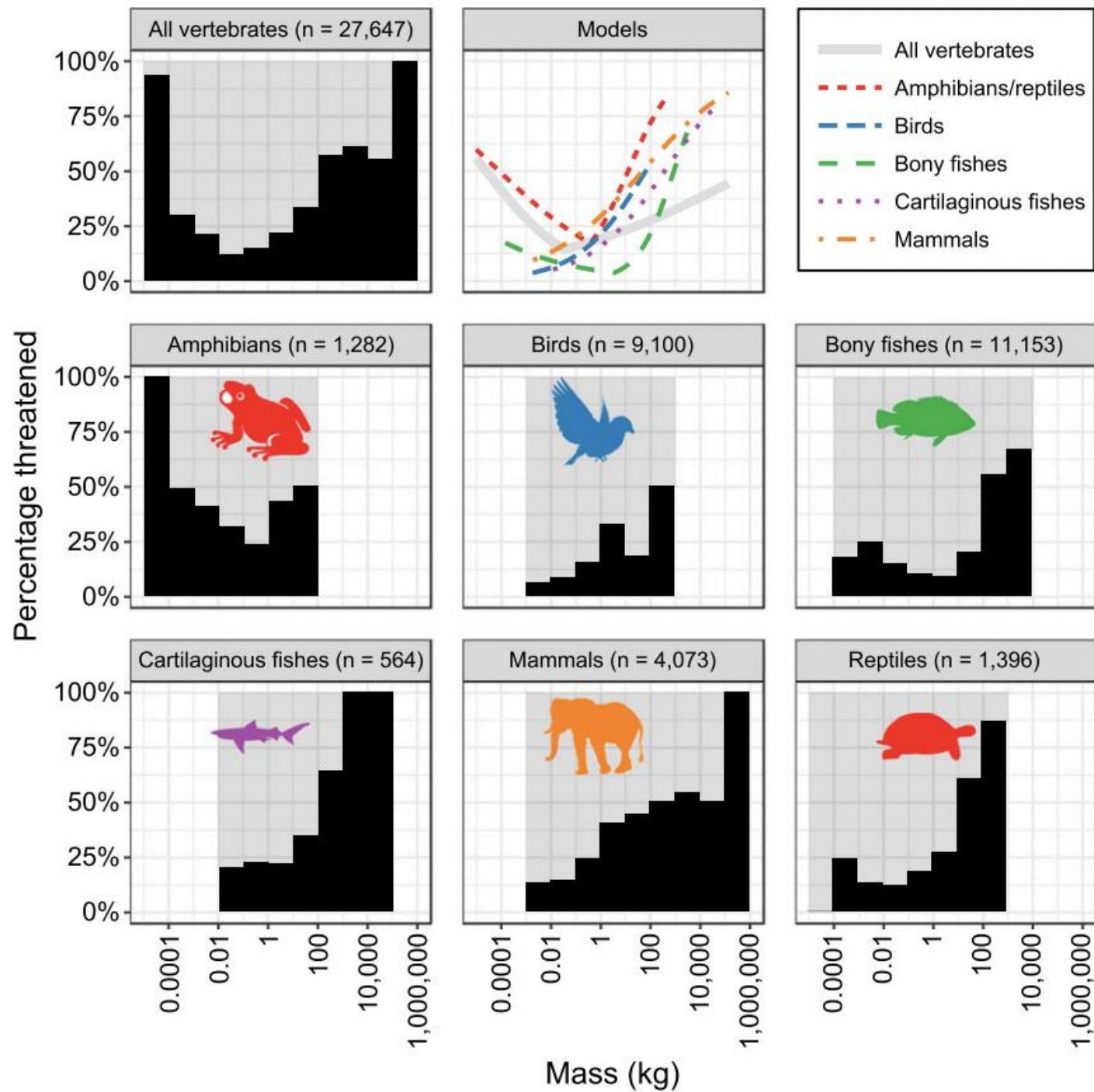
Palm Islands, Australia











NEXT CLASS

Monday, 04/07: Selection